

Chapter One

LORENTZ

TRANSFORMATION

1.1 Inertial Frame of Reference

Any frame can be denoted by a coordinate system with origin O and unit vector $\hat{i}$, $\hat{j}$ and $\hat{k}$ along direction x , y and z respectively as shown in figure.

If we are defining two different frame which means two different coordinate system.

The frame which is moving with constant velocity identified as inertial frame of reference. From any inertial frame of reference law of nature must be invariant. An inertial frame is also defined in which Newton's law of motion is defined.

Galilean Transformations:

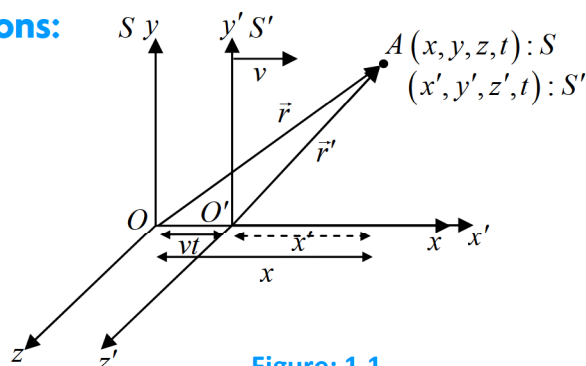


Figure: 1.1

There are two inertial frames denoted by S with origin O and S' with origin O' as shown in figure. At time $t = 0$ origin of both frames coincides. A frame S' which is moving with constant velocity v relative to an inertial frame S . Frame S' moving along $\hat{i}$ direction with respect to S . After time t the distance between O and O' is vt .

There is point A in space. After time t the position vector of A from origin O and O' is measured as $\vec{r}$ and $\vec{r}'$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\vec{r}' = x'\hat{i} + y'\hat{j} + z'\hat{k}$ respectively.

From the geometry $x = x' + vt$, $y = y'$ and $z = z'$.

We can write $x' = x - vt$, $y' = y$ and $z' = z$ which is linear transformation of co-ordinates from one inertial frame to another are referred as Galilean transformations.

Inverse Galilean transformation can be achieved just by changing the sign of velocity v to $-v$ given by

$$x = x' + vt, \quad y = y', \quad z = z',$$

In term of vector we can represent Galilean transformation and Inverse Galilean transformation as $\vec{r}' = \vec{r} - \vec{v}t$ and $\vec{r} = \vec{r}' + \vec{v}t$

Relative Velocity in Galilean Transformation

Let us assume point A is moving with respect to S' is u'_x, u'_y and u'_z in $\hat{i}, \hat{j}$ and $\hat{k}$ direction

where $u'_x = \frac{dx'}{dt}$, $u'_y = \frac{dy'}{dt}$ and $u'_z = \frac{dz'}{dt}$

The same point A is moving with respect to S with respect to u_x, u_y and u_z in $\hat{i}, \hat{j}$ and $\hat{k}$

direction where $u_x = \frac{dx}{dt}$, $u_y = \frac{dy}{dt}$ and $u_z = \frac{dz}{dt}$

The relation ship between speed measured by S and S' by differentiating with respect to time

t to equation $x = x' + vt$, $y = y'$, $z = z'$,

$$\frac{dx}{dt} = \frac{dx'}{dt} + v \Rightarrow u_x = u'_x + v, \quad \frac{dy}{dt} = \frac{dy'}{dt} \Rightarrow u_y = u'_y \quad \text{and} \quad \frac{dz}{dt} = \frac{dz'}{dt} \Rightarrow u_z = u'_z$$

Now we can have formula

$$u_x = u'_x + v \quad \text{or} \quad u'_x = u_x - v$$

$$u_y = u'_y, \quad u_z = u'_z$$

In the vector form relative velocity or velocity transformation is given as

$$\frac{d\vec{r}}{dt} = \vec{v} + \frac{d\vec{r}'}{dt}, \quad \vec{u} = \vec{v} + \vec{u}'$$

The Acceleration Transformation

Assume point A is moving with acceleration $\vec{a}' = \frac{d^2\vec{r}'}{dt^2}$ from frame S' . If observer will be

attached to frame S then he will measure acceleration of $\vec{a} = \frac{d^2\vec{r}}{dt^2}$.

Now let us find relation between $\vec{a}$ and $\vec{a}'$

Differentiate with respect to time t $\vec{r} = \vec{r}' + \vec{v}t$ we get $\frac{d\vec{r}}{dt} = \frac{d\vec{r}'}{dt} + \vec{v}$

Again differentiate $\frac{d\vec{r}}{dt} = \frac{d\vec{r}'}{dt} + \vec{v}$ with respect to time t we get $\frac{d^2\vec{r}}{dt^2} = \frac{d^2\vec{r}'}{dt^2} + \frac{d\vec{v}}{dt}$

$\vec{v}$ is constant so $\frac{d\vec{v}}{dt} = 0$, therefore, $\frac{d^2\vec{r}}{dt^2} = \frac{d^2\vec{r}'}{dt^2} \Rightarrow a = a'$

It is found that acceleration measured on both frames is same.

Inertial frame is also known as frame where Newton's law of motion is applicable.

Discrepancy in Galilean Transformation

Let us assume the light pulse is emerged from origin of frame S' and move along $\hat{i}$ direction.

The observer attached with S' frame will measure speed of light as $c = 3 \times 10^8 \text{ m/sec}$ in $\hat{i}$ direction.

If the observer will measure the speed of same light from S frame and if he will use Galilean transformation then he use formula $u_x = u'_x + v, u_y = u'_y, u_z = u'_z$, where $u'_x = c, u'_y = 0, u'_z = 0$. So measurement of velocity components of light from S is $u_x = c + v, u_y = 0$ and $u_z = 0$.

So we have seen that speed of light from S as $u' = c$, and same light will move from S' is with speed $u = c + v$ where $c = 3 \times 10^8 \text{ m/sec}$ is speed of light.

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1.2 Michelson Morley and Lorentz Transformation

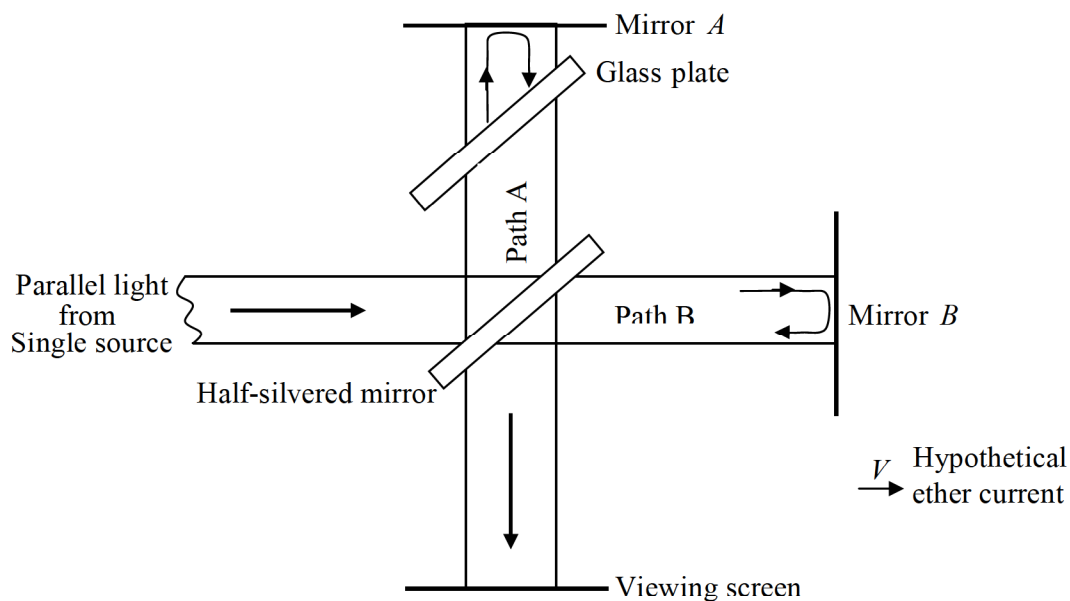


Figure 1.2

Albert A. Michelson (1852–1931) was born in Germany but came to the United States at the age of two with his parents, who settled in Nevada. He attended the U.S. Naval academy at Annapolis where, after two years of sea duty, he became a science instructor.

To improve his knowledge of optics, in which he wanted to specialize, Michelson went to Europe and studied in Berlin and Paris. Then he left the Navy to work first at the Case School of Applied Science in Ohio, then at Clark University in Massachusetts, and finally at the University of Chicago, where he headed the physics department from 1892 to 1929. Michelson's specialty was high-precision measurement, and for many decades his successive figures for the speed of light were the best available. He redefined the meter in terms of wavelengths of a particular spectral line and devised an interferometer that could determine the diameter of a star (stars appear as points of light in even the most powerful telescopes). Michelson's most significant achievement, carried out in 1887 in collaboration with Edward Morley, was an experiment to measure the motion of the earth through the "ether," a hypothetical medium pervading the universe in which light waves were supposed to occur. The notion of the ether was a hang-over from the days before light waves were recognized as electromagnetic, but nobody at the time seemed willing to discard the idea that light propagates relative to some sort of universal frame of reference.

To look for the earth's motion through the ether, Michelson and Morley used a pair of light beams formed by a half-silvered mirror, as in figure 1.2. One light beam is directed to a mirror along a path perpendicular to the ether current, and the other goes to a mirror along a path parallel to the ether current. Both beams end up at the same viewing screen. The clear glass plate ensures that both beams pass through the same thicknesses of air and glass. If the transit times of the two beams are the same, they will arrive at the screen in phase and will interfere constructively. An ether current due to the earth's motion parallel to one of the beams, however, would cause the beams to have different transit times and the result would be destructive interference at the screen. This is the essence of the experiment.

Although the experiment was sensitive enough to detect the expected ether drift, to everyone's surprise none was found. The negative result had two consequences. First, it showed that the ether does not exist and so there is no such thing as "absolute motion" relative to the ether: all motion is relative to a specified frame of reference, not to a universal one. Second, the result showed that the speed of light is the same for all observers, which is not true of waves that need a material medium in which to occur (such as sound and water waves). The Michelson-

Morley experiment set the stage for Einstein's 1905 special theory of relativity, a theory that Michelson himself was reluctant to accept. Indeed, not long before the flowering of relativity and quantum theory revolutionized physics, Michelson announced that "physical discoveries in the future are a matter of the sixth decimal place." This was a common opinion of the time. Michelson received a Nobel Prize in 1907, the first American to do so.

From Michelson Morley Experiment it is concluded that speed of light is invariant under any Inertial frame. So, there is Discrepancy that Galilean transformation is not adequate for any speed which will approach speed of light.

Lorentz Transformation

Postulates of Special Theory of Relativity

- (a) There is no universal frame of reference pervading all of space, so there is no such thing as "absolute motion".
- (b) The law of physics is the same in all frames of reference moving at constant velocity.
- (c) The speed of light in free space has the same value for all inertial observers. Any matter can not achieve speed of light i.e., $c = 3 \times 10^8 \text{ m/sec}$.

Hypothesis of Lorentz Transformation

- (a) It is linear in x and x' , so that a single event in frame S corresponds to a single event in frame S' .
- (b) For lower velocity it reduces to Galilean transformation.
- (c) If Lorentz transformation exist, then inverse transformation must exist by changing the sign of frame.

Derivation of Lorentz Transformation

Let us assume there is point in space A having space coordinates x, y, z and time coordinate t from inertial frame S . If same point A is measured from S' frame then space coordinates are x', y', z' and time t' .

According to hypothesis $x' = \gamma(x - vt)$

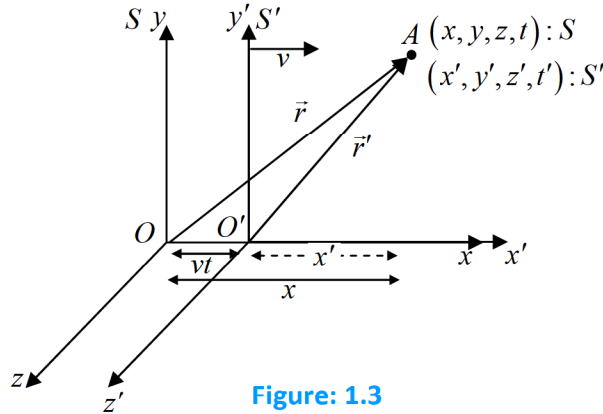


Figure: 1.3

Hence inverse Lorentz must exist then $x = \gamma(x' + vt')$, $y' = y$ and $z' = z$. The value of γ is same in both reference frame since only difference between S and S' is change of sign of velocity of reference frame v .

Put the value of $x' = \gamma(x - vt)$ in $x = \gamma(x' + vt')$, one will get $x = \gamma^2(x - vt) + \gamma vt'$

and $t' = \gamma t + \left(\frac{1 - \gamma^2}{\gamma v} \right) x$.

It is assumed that at $t = t' = 0$ the origin of both S and S' is same. We assume that light pulse emerged at $t = t' = 0$ and moves in $\hat{i}$ direction.

From S frame the relation between position and time is $x = ct$ in S frame and $x' = ct'$ in S' frame.

Use relation $x' = ct'$ put the value of x' and t' as function of x and t .

Solve in another way

Method 1: Multiply $x' = \gamma(x - vt)$ to $x = \gamma(x' + vt')$

$$x'x = \gamma^2(x - vt)(x' + vt') \Rightarrow x'x = \gamma^2(xx' - vxt' + vtx' - v^2t't)$$

Divide both side by xx' so we get

$$1 = \gamma^2 \left(1 - \frac{vt}{x} + \frac{vt'}{x'} - v^2 \frac{t}{x} \cdot \frac{t'}{x'} \right) \Rightarrow 1 = \gamma^2 \left(1 - \frac{v}{c} + \frac{v}{c} - \frac{v^2}{c^2} \right) \Rightarrow \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Method 2: From S frame the relation between position and time is $x = ct$ in S frame and $x' = ct'$ in S' frame.

Use relation $x' = ct'$ put the value of x' and t' as function of x and t .

Solve in another way

$$\gamma(x - vt) = c\gamma t + \left(\frac{1 - \gamma^2}{\gamma v}\right) cx$$

$$x \left(\gamma - \left(\frac{1 - \gamma^2}{\gamma v} \right) c \right) = ct \left(\gamma + \frac{\gamma v}{c} \right) \Rightarrow x = ct \frac{\gamma + \frac{\gamma v}{c}}{\gamma - \left(\frac{1 - \gamma^2}{\gamma v} \right) c}$$

Compare above equation with $x = ct$ we get

$$\frac{\gamma + \frac{\gamma v}{c}}{\gamma - \left(\frac{1 - \gamma^2}{\gamma v} \right) c} = 1 \Rightarrow \gamma + \frac{\gamma v}{c} = \gamma - \left(\frac{1 - \gamma^2}{\gamma v} \right) c \Rightarrow \frac{\gamma^2 v^2}{c^2} = \gamma^2 - 1 \Rightarrow \gamma^2 = 1 - \frac{v^2}{c^2} \Rightarrow \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Put the value of $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$ in equation $x' = \gamma(x - vt)$ and $t' = \gamma t + \left(\frac{1 - \gamma^2}{\gamma v} \right) x$ we get

Lorentz transformation and inverse Lorentz transformation as

Lorentz transformation	Inverse Lorentz transformation
$x' = \gamma(x - vt) = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}}$	$x = \gamma(x' + vt') = \frac{x' + vt'}{\sqrt{1 - \frac{v^2}{c^2}}}$
$y' = y$	$y = y'$
$z' = z$	$z = z'$
$t' = \gamma \left(t - \frac{vx}{c^2} \right) = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$	$t = \gamma \left(t' + \frac{vx'}{c^2} \right) = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$

Example: Two events in S frame is separated by space coordinate $x_2 - x_1 = L$ and by time coordinate $t_2 - t_1 = T$.

(a) What will be difference in space coordinate measured from S' which is moving with speed v with respect to S frame.

(b) What will be difference in time coordinate measured from S' which is moving with speed v with respect to S frame.

Solution: From S frame $x_2 - x_1 = L$ and $t_2 - t_1 = T$ from S' frame $x'_2 - x'_1 = ?$ and $t'_2 - t'_1 = ?$

$$(a) L' = x'_2 - x'_1 = \left(\frac{x_2 - x_1}{\sqrt{1 - \frac{v^2}{c^2}}} \right) - \left[\frac{v(t_2 - t_1)}{\sqrt{1 - \frac{v^2}{c^2}}} \right] = \frac{L - vT}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$(b) T' = t'_2 - t'_1 = \frac{t_2 - \frac{vx_2}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{t_1 - \frac{vx_1}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{t_2 - t_1}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{v}{c^2} \frac{x_2 - x_1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{T}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{v}{c^2} \frac{L}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Example: A relativistic particle moves with a constant velocity v with respect to the laboratory frame. In time τ , measured in the rest frame of the particle,

(a) What will distance travelled by the particle in its own reference frame.

(b) If τ is time measured by particle in particle frame what is time measured in laboratory frame.

(c) What is the distance that it travels in the laboratory frame.

Solution: Assume laboratory is S frame, the particle is moving with respect to laboratory so it is S' frame.

(a) From Particle frame of reference

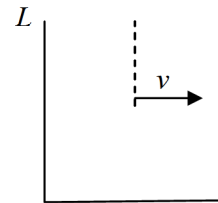
$$x'_1 = 0, x'_2 = 0 \quad t_{initial} = t'_1, t_{final} = t'_2 \Rightarrow t'_2 - t'_1 = \tau$$

$$d = x'_2 - x'_1 = 0$$

$$(b) t_2 - t_1 = \frac{t'_2 - \frac{vx'_2}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{t'_1 - \frac{vx'_1}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{t'_2 - t'_1}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{v}{c^2} \left(\frac{x'_2 - x'_1}{\sqrt{1 - \frac{v^2}{c^2}}} \right) = \frac{\tau}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$(c) x_1 = \frac{x'_1 + vt'_1}{\sqrt{1 - v^2/c^2}}, x_2 = \frac{x'_2 + vt'_2}{\sqrt{1 - v^2/c^2}}$$

$$x_2 - x_1 = \frac{x'_2 - x'_1}{\sqrt{1 - v^2/c^2}} + \frac{v(t'_2 - t'_1)}{\sqrt{1 - v^2/c^2}} \Rightarrow \Delta x = \frac{v(t'_2 - t'_1)}{\sqrt{1 - v^2/c^2}} = \frac{v\tau}{\sqrt{1 - v^2/c^2}}$$



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1.3 Relative Velocity

Let the co-ordinates of a particle in frame S be (x, y, z, t) and in frame $S'(x', y', z', t')$ then the component of its velocity in two frame can be written as

There is one inertial frame S , S' is another inertial frame moving with respect to S in x -direction and A is another inertial frame which is moving with respect to S' with velocity component (u'_x, u'_y, u'_z) .

$$\text{So, } u'_x = \frac{dx'}{dt'}, u'_y = \frac{dy'}{dt'}, u'_z = \frac{dz'}{dt'}$$

The velocity component of A from S frame is given by

$$u_x = \frac{dx}{dt}, u_y = \frac{dy}{dt}, u_z = \frac{dz}{dt}$$

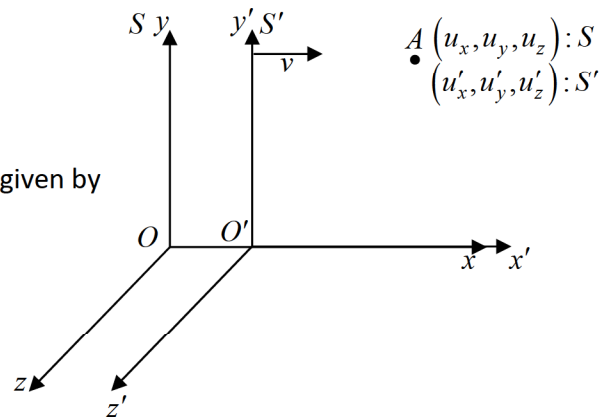


Figure: 1.4

From the inverse Lorentz Transformation,

$$x = \gamma(x' + vt') \text{ , } y = y' \text{ , } z = z' \text{ , } t = \gamma\left(t' + \frac{vx'}{c^2}\right) \text{ where } \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

differentiating both sides, we get

$$dx = \gamma(dx' + vdt') \text{ , } dy = dy' \text{ , } dz = dz' \text{ and } dt = \gamma\left(dt' + \frac{vdx'}{c^2}\right)$$

$$u_x = \frac{dx}{dt} = \frac{\gamma}{\gamma} \frac{dx' + vdt'}{dt' + \frac{vdx'}{c^2}} = \frac{\frac{dx'}{dt'} + v}{1 + \frac{v}{c^2} \frac{dx'}{dt'}} \Rightarrow u_x = \frac{u'_x + v}{1 + \frac{v}{c^2} u'_x}$$

$$u_y = \frac{dy}{dt} = \frac{dy'}{\gamma\left(dt' + \frac{vdx'}{c^2}\right)} = \frac{\frac{dy'}{dt'}}{\gamma\left(1 + \frac{v}{c^2} \frac{dx'}{dt'}\right)} \Rightarrow u_y = \frac{u'_y \sqrt{1 - \frac{v^2}{c^2}}}{\left(1 + \frac{vu'_x}{c^2}\right)}$$

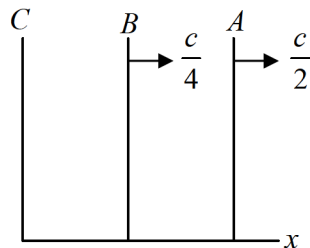
$$u_z = \frac{dz}{dt} = \frac{dz'}{\gamma\left(dt' + \frac{vdx'}{c^2}\right)} = \frac{u'_z \sqrt{1 - \frac{v^2}{c^2}}}{\left(1 + \frac{v}{c^2} \frac{dx'}{dt'}\right)} \Rightarrow u_z = \frac{u'_z \sqrt{1 - \frac{v^2}{c^2}}}{\left(1 + \frac{vu'_x}{c^2}\right)}$$

From Lorentz transformation we can get $u'_x = \frac{u_x - v}{1 - \frac{vu_x}{c^2}}$, $u'_y = \frac{u_y \sqrt{1 - \frac{v^2}{c^2}}}{\left(1 - \frac{vu_x}{c^2}\right)}$, $u'_z = \frac{u_z \sqrt{1 - \frac{v^2}{c^2}}}{\left(1 - \frac{vu_x}{c^2}\right)}$

Example: A frame A move with respect to frame B with speed $\frac{c}{2}$ in positive $\hat{x}$

direction .Another frame C moves with respect to B with speed $\frac{c}{4}$ in positive $\hat{x}$ with speed $\frac{c}{4}$.

What will speed of C with respect to observer attached to frame A ?



Solution: It is given $v_{B,A} = \frac{c}{2}$ and $v_{C,B} = \frac{c}{4}$

We assume frame A is equivalent to S then frame B as S'

The given information is $v_{B,A} = v_{S',S} = v = \frac{c}{2}\hat{i}$ $v_{C,B} = v_{C,S'} = u'_x = \frac{c}{4}$ and $v_{C,A} = u_x = ?$

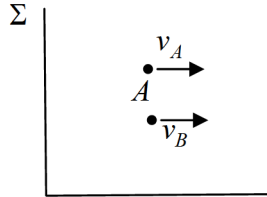
So we use formula $u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{\frac{c}{4} + \frac{c}{2}}{1 + \frac{\frac{c}{4} \cdot \frac{c}{2}}{c^2}} = \frac{\frac{c+2c}{4}}{1 + \frac{1}{8}} = \frac{\frac{3c}{4}}{\frac{9}{8}} = \frac{2}{3}c$

Example: Two particles A, B moving with respect to common frame Σ . If speed of A, B is measured by observer attached to Σ are v_A and v_B in positive $\hat{x}$ direction respectively.

(a) By using Lorentz transformation Find the speed of B with respect to observer attached to A .

(b) Find the speed of B with respect to observer attached to A if $\frac{v_B}{c} \rightarrow 0$

(c) Find the speed of B with respect to observer attached to A if $v_B = c$



Solution:

(a) $v_{A,\Sigma} = v_A\hat{i}$ $v_{B,\Sigma} = v_B\hat{i}$ $v_{B,A} = ?$

Assume particle A as S so Σ is identified as S'

So $v_{\Sigma,A} = v_{S',S} = v = -v_A\hat{i}$, $v_{B,\Sigma} = v_{B,S'} = u'_x = v_B\hat{i}$ and $v_{B,A} = v_{B,S} = u_x = ?$

$$v_{B,A} = u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{v_B + (-v_A)}{1 + \frac{v_B \cdot (-v_A)}{c^2}} = \frac{v_B - v_A}{1 - \frac{v_B \cdot v_A}{c^2}}$$

(b) $v_{B,A} = u_x = \frac{v_B - v_A}{1 - \frac{v_B \cdot v_A}{c^2}}$ if $\frac{v_B}{c} \rightarrow 0$ $u_x = v_B - v_A$, which is exactly result obtain by Galilean

transformation

$$(c) \quad v_{B,A} = u_x = \frac{v_B - v_A}{1 - \frac{v_B \cdot v_A}{c^2}} \quad v_B = c$$

$$v_{B,A} = u_x = \frac{c - v_A}{1 - \frac{c \cdot v_A}{c^2}} = \left(\frac{c - v_A}{c - v_A} \right) c = c, \text{ which endorsed the postulates that speed of light is}$$

invariant from inertial frame of reference.

Example: Two particles A, B moving with respect to common inertial frame Σ . If speed of A, B is measured by observer attached to Σ are v_A and v_B in positive $\hat{x}$ and $\hat{y}$ direction respectively.

(a) By using Lorentz transformation Find the velocity of B with respect to observer attached to A .

(b) Find the speed of B with respect to observer attached to A if $v_B = c$

$$v_{A,\Sigma} = v_A \hat{i} \quad v_{B,\Sigma} = v_B \hat{j} \quad v_{B,A} = ?$$

Solution:

(a) Assume particle A as S so Σ is identified as S'

So $v_{\Sigma,A} = v_{S',S} = v = -v_A \hat{i}$, $\vec{v}_{B,\Sigma} = \vec{v}_{B,S'} = u'_x = 0, u'_y = v_B, u'_z = 0$ and $\vec{v}_{B,A} = \vec{v}_{B,S} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k} = ?$

$$u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{0 + (-v_A)}{1 + \frac{0 \cdot (-v_A)}{c^2}} = \frac{0 - v_A}{1 - \frac{0 v_A}{c^2}} = -v_A, \quad u_y = \frac{u'_y \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{u'_x v}{c^2}} = \frac{v_B \sqrt{1 - \frac{(-v_A)^2}{c^2}}}{1 + \frac{0 \cdot (-v_A)}{c^2}} = v_B \sqrt{1 - \frac{v_A^2}{c^2}}$$

$$\text{Similarly, } u_z = \frac{u'_z \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{u'_x v}{c^2}} = \frac{0 \sqrt{1 - \frac{(-v_A)^2}{c^2}}}{1 + \frac{0 \cdot (-v_A)}{c^2}} = 0, \quad \vec{v}_{B,A} = \vec{v}_{B,S} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k} = -v_A \hat{i} + v_B \sqrt{1 - \frac{v_A^2}{c^2}} \hat{j}$$

$$(b) \text{ For } v_B = c \quad \vec{v}_{B,A} = -v_A \hat{i} + c \sqrt{1 - \frac{v_A^2}{c^2}} \hat{j}$$

$$|v_{B,A}| = \sqrt{u_x^2 + u_y^2 + u_z^2} = \sqrt{v_A^2 + c^2 \left(1 - \frac{v_A^2}{c^2} \right)} = c, \text{ again we have seen the speed of light is invariant}$$

from any inertial frame.

Example: A spaceship moving away from the earth with velocity $0.5c$ fires a rocket whose velocity relative to space is $0.5c$ (a) Away from earth (b) Towards the earth. Calculate velocity of the rocket as observed from the earth in two cases.

Solution: (a) Assume that spaceship moving with respect to Earth in positive $\hat{x}$ direction.

So velocity of space ship with respect to Earth is $v_{SP,E} = 0.5$, $v_{R,SP} = 0.5c$ away from earth $v_{R,E} = ?$

Assume Earth as S frame, then space ship is S' frame and speed of Rocket is given with respect to spaceship i.e. S' .

$$v_{SP,E} = v_{S',S} = v = 0.5c, v_{R,SP} = v_{R,S'} = u'_x = 0.5c$$

$$(a) u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{0.5c + 0.5c}{1 + \frac{0.5c \times 0.5c}{c^2}} = \frac{c}{1 + 0.25} = \frac{c}{1.25} = \frac{4}{5}c \Rightarrow u = 0.8c$$

(b) $v_{SP,E} = v_{S',S} = v = 0.5c$, $v_{R,SP} = v_{R,S'} = u'_x = -0.5c$ rocket is moving towards earth $v_{R,E} = ?$

$$u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{-0.5c + 0.5c}{1 + \frac{-0.5c \times 0.5c}{c^2}} = 0$$

Example: A glass plate of refractive index μ moving with speed v in positive $\hat{x}$ - direction with respect to laboratory. Light pulse is entered into glass plate along $\hat{x}$ - axis.

(a) Find the speed of light in the glass plate with respect to observer attached to Glass plate.

(b) Find the speed of light in the glass plate with respect to observer attached to Laboratory

Solution: (a) If observer is attached to glass plate then glass plate is rest with respect to observer then speed of light in the glass plate is $v_{l,g} = \frac{c}{\mu}$.

(b) If observer is attached to laboratory then $v_{g,L} = v\hat{x}$, $v_{l,g} = \frac{c}{\mu}$

We assume laboratory as S frame then glass plate is treated as S' so $v_{g,L} = v_{S',S} = v\hat{x}$,

so $v_{l,g} = v_{l,S'} = u'_x = \frac{c}{\mu}$, $v_{l,L} = v_{l,S} = u_x = ?$

$$u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{\frac{c}{\mu} + v}{1 + \frac{\frac{c}{\mu} \times v}{c^2}} = \left(\frac{c + \mu v}{\mu c + v} \right) c$$

Example: A light pulse emerge from origin of frame A . If observer is attached to frame the light ray is appeared to make angle θ_0 with x axis and movement of light is confine in $x-y$ plane. Frame A is moving with respect to frame B with speed v in positive x -axis. If observer will be attached with frame B , what will velocity of light pulse?

Solution: Assume frame B is S frame, Then frame A is treated as S' so $v_{S',S} = v \hat{x}$

Speed of light with respect to frame A in $\hat{x}$ direction is identified as $u'_x = c \cos \theta_0$

Speed of light with respect to frame A in $\hat{y}$ direction is identified as $u'_y = c \sin \theta_0$

Speed of light with respect to frame A in $\hat{z}$ direction is identified as $u'_z = 0$

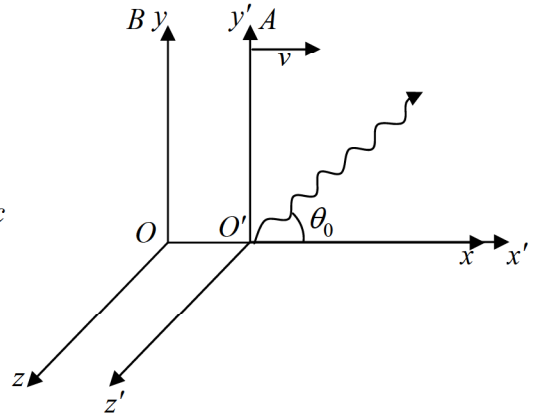
We need to find velocity of light with respect to S frame i.e. $u_x = ?$, $u_y = ?$, $u_z = ?$

$$u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{c \cos \theta_0 + v}{1 + \frac{c \cos \theta_0 v}{c^2}} = \left(\frac{c \cos \theta_0 + v}{c + v \cos \theta_0} \right) c$$

$$u_y = \frac{u'_y \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{u'_x v}{c^2}} = \frac{c \sin \theta_0 \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{c \cos \theta_0 v}{c^2}} = \left(\frac{c \sin \theta_0 \sqrt{1 - \frac{v^2}{c^2}}}{c + v \cos \theta_0} \right) c$$

$$u_z = \frac{u'_z \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{u'_x v}{c^2}} = \frac{0 \times \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{c \cos \theta_0 v}{c^2}} = 0$$

$$\vec{v}_{l,B} = \vec{u} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k} = \left(\frac{c \cos \theta_0 + v}{c + v \cos \theta_0} \right) c \hat{i} + \left(\frac{c \sin \theta_0 \sqrt{1 - \frac{v^2}{c^2}}}{c + v \cos \theta_0} \right) c \hat{j}$$



Chapter One

LORENTZ

TRANSFORMATION

1.4 Consequences of Lorentz Transformation (Length Contraction, Time Dilation and Loss of Simultaneity)

(a) Length Contraction:

Moving rod appeared shorter

Proper Length: If relative speed between length of rod and observer is zero then observer will measure proper length.

Contracted Length: If there is relative speed between length of rod and observer then observer will measure contracted length which is shorter than proper length.

Suppose a rod of proper length l_0 lying along $\hat{x}$ axis of S' frame which is moving with speed v with respect to S along $\hat{x}$ direction.

In order to measure the length of an object in motion, relation to observer, the position of two end points recorded simultaneously, the length of object in direction of motion appeared smaller to observer.

If observer is attached to S' he will measure

proper length i.e. $l_0 = x'_2 - x'_1$

If observer is attached to S he will measure proper

length i.e. $l = x_2 - x_1$

From Lorentz transformation

$$x'_2 - x'_1 = \gamma(x_2 - vt_2) - \gamma(x_1 - vt_1)$$

$$(x'_2 - x'_1) = \gamma(x_2 - x_1) \text{ because } t_2 = t_1$$

$$(x'_2 - x'_1) = \frac{x_2 - x_1}{\sqrt{1 - v^2/c^2}} \Rightarrow (x_2 - x_1) = \sqrt{1 - \frac{v^2}{c^2}} (x'_2 - x'_1)$$

$$\Rightarrow l = l_0 \sqrt{1 - \frac{v^2}{c^2}} \quad \text{so } l < l_0$$

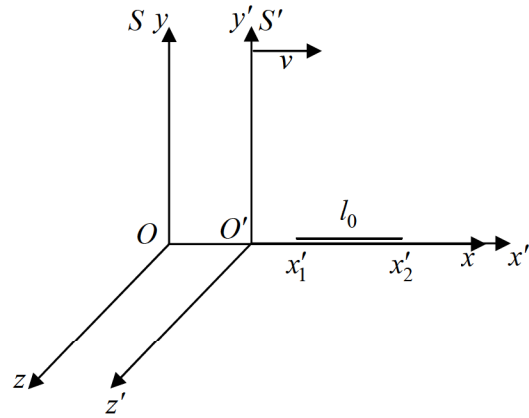


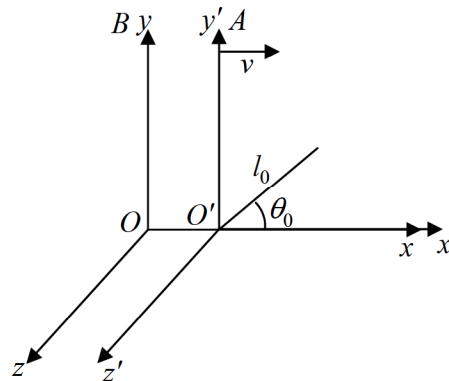
Figure 1.5

$l < l_0$, which means moving rod appeared shorter, such a contraction of length in direction of motion relative to observer is called Lorentz Fitzgerald contradiction.

Example: A rod of proper length l_0 making angle θ_0 with x -axis of frame A . If frame A is moving with respect observer along x -axis.

(a) Determine the length of rod measured by observer.

(b) What will angle made by rod with respect to observer as measured by observer.



Solution: Proper length of the rod in the direction of moving frame, $l_{x_0} = l_0 \cos \theta_0$, and perpendicular to motion is $l_{y_0} = l_0 \sin \theta_0$

(a) Length measured in the moving frame $l_x = l_0 \cos \theta_0 \sqrt{1 - \frac{v^2}{c^2}}$ and $l_y = l_0 \sin \theta_0$

$$|l| = \sqrt{l_0^2 \cos^2 \theta_0 \left(1 - \frac{v^2}{c^2}\right) + l_0^2 \sin^2 \theta_0} = l_0 \sqrt{\cos^2 \theta_0 \left(1 - \frac{v^2}{c^2}\right) + \sin^2 \theta_0} = l_0 \sqrt{1 - \left(\frac{v \cos \theta_0}{c}\right)^2}$$

$$(b) \tan \theta' = \frac{l_y}{l_x} = \frac{l_0 \sin \theta_0}{l_0 \cos \theta_0 \sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow \tan \theta' = \gamma \tan \theta_0, \theta' = \tan^{-1}(\gamma \tan \theta_0)$$

Example: The area of a disc in its rest frame S is equal to 1 (in some units). The disc will appear distorted to an observer O moving with a speed u with respect to S along the plane of the disc. Find the area of the disc measured in the rest frame of the observer O is (c is the speed of light in vacuum)

Solution: Area of disc from S - frame is 1 i.e., $\pi a^2 = 1$ or $\pi a \cdot a = 1$

Due to length contraction, the circle will appear as ellipse with major axis a and minor axis

$$b = a \sqrt{1 - \frac{u^2}{c^2}}$$

$$\text{Area of disc from } S' \text{ frame is } \pi a \cdot b = \pi a \cdot a \sqrt{1 - \frac{u^2}{c^2}} = 1 \cdot \sqrt{1 - \frac{u^2}{c^2}} = \sqrt{1 - \frac{u^2}{c^2}}$$

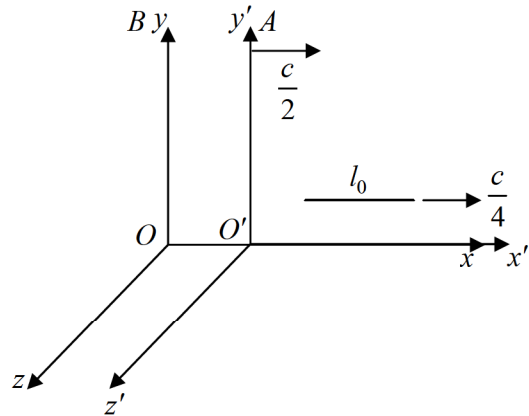
where u is the speed of circle with respect to observer.

Example: A rod of proper length l_0 moving with respect to A with speed velocity $\frac{c}{4} \hat{i}$. The frame

A is moving with respect to B with velocity $\frac{c}{4} \hat{i}$

(a) Find length of rod measured by observer attached to rest frame of A

(b) Find length of rod measured by observer attached to rest frame of B



Solution: (a) velocity of rod with respect to A $v_{R,A} = \frac{c}{4}$

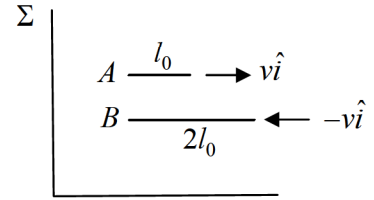
$$l = l_0 \sqrt{1 - \frac{(v_{R,A})^2}{c^2}} = l_0 \sqrt{1 - \frac{1}{16}} = \sqrt{\frac{15}{16}} l_0$$

(b) velocity of rod with respect to A $v_{R,A} = v_{S',A} = u'_x = \frac{c}{4}$

$$\text{velocity of } v_{A,B} = v_{S',S} = v = \frac{c}{2} \quad v_{R,B} = v_{R,S} = u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{\frac{c}{2} + \frac{c}{4}}{1 + \frac{\frac{c}{2} \cdot \frac{c}{4}}{c^2}}$$

$$l = l_0 \sqrt{1 - \frac{(v_{R,B})^2}{c^2}} = l_0 \sqrt{1 - \frac{4}{9}} = \sqrt{\frac{5}{9}} l_0$$

Example: Two rod A and B moving with respect to common frame Σ with velocity $v\hat{i}$ and $-v\hat{i}$ respectively. The proper length of A is l_0 and proper length of B is $2l_0$. Find the value of v such that length of rod B will appeared l_0 from the observer is attached to rest frame of A .



Solution: $v_{A,\Sigma} = v\hat{i}$, $v_{B,\Sigma} = -v\hat{i}$

First, we need to calculate $v_{B,A}$

$$A \equiv S, \Sigma \equiv S' \quad v_{S',S} = -v\hat{i} \quad v_{B,\Sigma} = v_{B,S'} = u'_x = -v\hat{i}$$

$$v_{B,A} = v_{B,S} = u_x \quad u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{-v - v}{1 + \frac{(-v)(-v)}{c^2}} = \frac{-2v}{1 + \frac{v^2}{c^2}}$$

$$l_0 = 2l_0 \sqrt{1 - \frac{u_x^2}{c^2}} = l_0 = 2l_0 = \frac{3}{4} = \frac{u_x^2}{c^2} \Rightarrow \frac{3}{4} = \frac{4 \frac{v^2}{c^2}}{\left(1 + \frac{v^2}{c^2}\right)^2}$$

put $\frac{v^2}{c^2} = \alpha < 1$

$$\frac{3}{4} = \frac{4\alpha}{(1+\alpha)^2} \Rightarrow 16\alpha = 3 + 3\alpha^2 + 6\alpha \Rightarrow 3\alpha^2 - 10\alpha + 3 = 0 \Rightarrow 3\alpha(\alpha - 3) - 1(\alpha - 3) = 0$$

So $\alpha = \frac{1}{3}$, $\alpha = 3$ the accepted solution is so $v = \frac{c}{\sqrt{3}}$

Chapter One

LORENTZ

TRANSFORMATION

1.4 Consequences of Lorentz Transformation

(b) Time Dilation:

Moving clock ticks slower

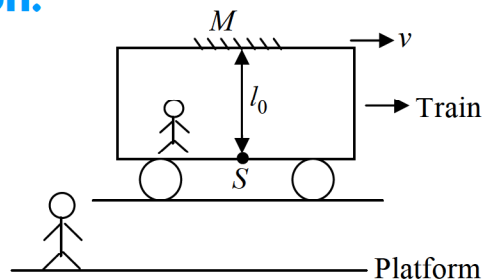


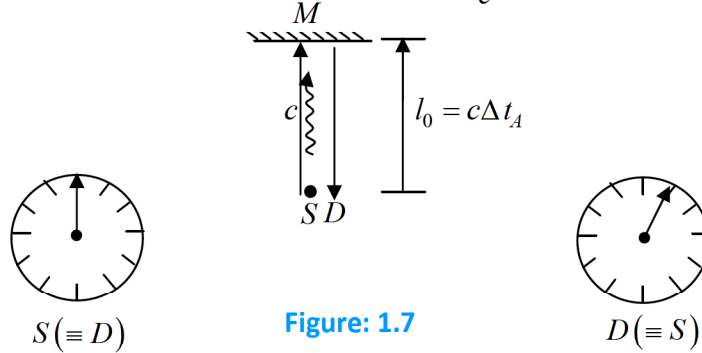
Figure: 1.6

Assume a train is moving with respect to platform. The train is moving with velocity v along positive x direction. The light source and detector is attached at the floor of train.

A mirror is attached roof of train which is just above the light source and detector. The height between roof and floor is l_0 . The experiment is defined as light pulse will emerge from source hit the mirror and again hit detector. There two observers one observer A is attached to train

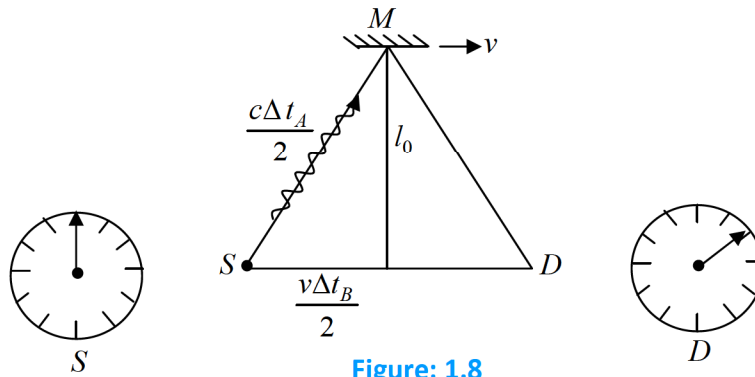
and another observer is B attached to platform. We need to find time interval such that light pulse will emerge from source and again detected by the detector.

From observer A point of view the time interval is $\Delta t_A = \frac{2l_0}{c} = \Delta t_0$.



From observer B point of view the time interval is Δt_B . From the geometry and using

Pythagoras law $l_0 = \sqrt{c^2 \left(\frac{\Delta t_B}{2} \right)^2 - v^2 \left(\frac{\Delta t_B}{2} \right)^2} \Rightarrow \Delta t_B = \frac{\frac{2l_0}{c}}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{\Delta t_A}{\sqrt{1 - \frac{v^2}{c^2}}}$



We can see $\Delta t_B > \Delta t_A$

Proper Time: If clock is rest with respect to observer then observer will observe proper time

Time Dilation: When two observers are in relative uniform motion and uninfluenced by any gravitational mass, the point of view of each will be that the other's (moving) clock is ticking at a *slower* rate than the local clock which will measure proper time. The faster the relative velocity, the greater the magnitude of time dilation. This case is sometimes called special relativistic time dilation.

Derivation of Time Dilation from Lorentz Transformation

A clock being at rest in the S' frame measures the time t'_2 and t'_1 of two events occurring at a fixed position x' , then time interval Δt measures from S frame appears slow (Δt_0) from S' frame i.e. to the observer the moving clock will appear to go slow.

$$\Delta t' = t'_2 - t'_1 = \Delta t_0 \Rightarrow \Delta t = t_2 - t_1 = \gamma \left(t'_2 + \frac{vx'_2}{c^2} \right) - \gamma \left(t'_1 + \frac{vx'_1}{c^2} \right) \Rightarrow \Delta t = \frac{t'_2 - t'_1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where Δt_0 proper time which is measured from rest frame of clock, and Δt is dilated time which is measured from moving frame of clock where $\Delta t > \Delta t_0$.

Example: A muon is having life time $2.2\mu\text{sec}$ moving towards earth. The speed of muon respect to earth is $0.998c$. Ignore the mass of muon find the distance travelled by muon with respect to observer at earth before it decays.

Solution: The proper life time is given as $t_0 = 2.2 \times 10^{-6} \text{ sec}$

The speed of muon with respect to observer is $v = 0.998c$

The clock on the muon is moving with respect to observer on earth so with respect to observer the life time is dilated and measured as

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{2.2 \times 10^{-6} \text{ sec}}{\sqrt{1 - \left(\frac{0.998c}{c} \right)^2}} = 34.6 \times 10^{-6} \text{ sec}.$$

Let the distance traveled by muon with respect to earth is d

$$d = vt = 0.998c \times 34 \times 10^{-6} \text{ m/sec} = 0.998 \times 3 \times 10^8 \times 34 \times 10^{-6} \text{ m/sec} = 10.4 \text{ km}.$$

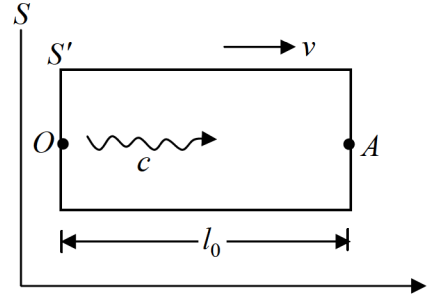
Example: A space craft is moving with respect to earth. As an observer on earth finds that between 1pm to 2pm according to her clock 3601 sec elapsed the on spacecraft clock .what is speed of space craft with respect to earth.

Solution: The proper time interval during 1pm to 2pm is $\Delta t_0 = 3600 \text{ sec}$

The clock attached to space craft is moving with respect to earth so it will be measured dilated time i.e. $\Delta t = 3601 \text{ sec}$

$$\text{So } \Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow \sqrt{1 - \frac{v^2}{c^2}} = \frac{\Delta t_0}{\Delta t} \Rightarrow v = c \sqrt{1 - \left(\frac{\Delta t_0}{\Delta t} \right)^2} = c \sqrt{1 - \left(\frac{3600}{3601} \right)^2} = 7.1 \times 10^6 \text{ m/sec}$$

Example: The distance between O to A is proper length l_0 in rest frame of S' as shown in figure, what is time measured by observer attached to rest frame of S such that light emerge from O to reach to A . It is given that velocity of S' with respect to S is v .



Solution:

Method 1: From frame S' $x'_2 - x'_1 = l_0, t'_2 - t'_1 = \frac{l_0}{c}$

$$\text{From frame } S \quad t_2 - t_1 = \left(\frac{t'_2 + v \frac{x'_2}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \right) - \left(\frac{t'_1 + v \frac{x'_1}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \right) = \left(\frac{t'_2 - t'_1}{\sqrt{1 - \frac{v^2}{c^2}}} \right) + \frac{v}{c^2 \sqrt{1 - \frac{v^2}{c^2}}} (x'_2 - x'_1)$$

$$t_2 - t_1 = \left(\frac{\frac{l_0}{c}}{\sqrt{1 - \frac{v^2}{c^2}}} \right) + \frac{v l_0}{c^2 \sqrt{1 - \frac{v^2}{c^2}}} = \frac{l_0}{c} \frac{1 + \frac{v}{c}}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{l_0}{c} \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}}$$

Method 2: From frame S

$$c \Delta t = l_0 \sqrt{1 - \frac{v^2}{c^2}} + v \Delta t \Rightarrow \Delta t = (c - v) \Delta t = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$\Delta t = \frac{l_0 \sqrt{1 - \frac{v^2}{c^2}}}{c - v} = \frac{l_0}{c} \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}}$$

Chapter One

LORENTZ

TRANSFORMATION

1.4 Consequences of Lorentz Transformation

(c) Loss of Simultaneity

The loss of simultaneity in Special relativity - is that real or created due to the fact that light takes time to travel. So even though two events are simultaneous but since light takes time to travel, they may not be simultaneous for two different observers.

The loss of simultaneity can be understood with following experiment.

A train of proper length l_0 moving with speed v with respect to a platform in positive $\hat{x}$ direction. There are two detectors A and B are attached at two opposite ends of the train. Two observers X and Y attached to train and platform respectively.

Suddenly two light pulse immerged from center of train O and moves in straight line towards detectors A and B . We

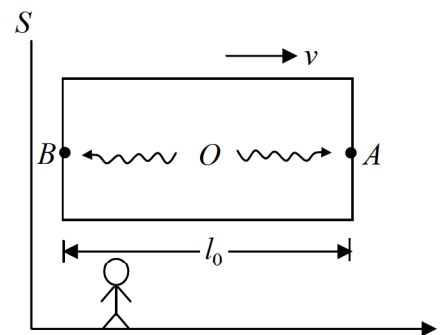


Figure: 1.6

need to calculate time interval such that light pulse is detected by A and B with respect to observer X and Y .

With respect to observer X :

Event-1: Let us assume that $\Delta t_{A,X}$ is time interval measured by observer X such that light pulse emerge from O and detected by detector A so $\Delta t_{A,X} = \frac{l_0}{2c}$

Event-2: Let us assume that $\Delta t_{B,X}$ is time interval measured by observer X such that light pulse emerge from O and detected by detector B so $\Delta t_{B,X} = \frac{l_0}{2c}$

We can see $\Delta t_{A,X} = \Delta t_{B,X}$ which means two events are simultaneous with respect to observer X

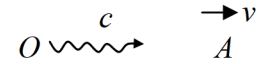
With respect to observer Y :

Event-1: Let us assume that $\Delta t_{A,Y}$ is time interval measured by observer Y such that light pulse emerge from O and detected by detector A .

Using kinematics relation

$$c\Delta t_{A,Y} = \frac{l_0}{2} \sqrt{1 - \frac{v^2}{c^2}} + v\Delta t_{A,Y} \Rightarrow (c - v)\Delta t_{A,Y} = \frac{l_0}{2} \sqrt{1 - \frac{v^2}{c^2}}$$

$$\Delta t_{A,Y} = \frac{l_0 \sqrt{1 - \frac{v^2}{c^2}}}{2(c - v)} = \frac{l_0}{2c} \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}}$$

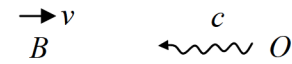


Event-2: Let us assume that $\Delta t_{B,Y}$ is time interval measured by observer Y such that light pulse emerge from O and detected by detector B so

Using kinematics relation

$$c\Delta t_{B,Y} = \frac{l_0}{2} \sqrt{1 - \frac{v^2}{c^2}} - v\Delta t_{B,Y} \Rightarrow (c + v)\Delta t_{B,Y} = \frac{l_0}{2} \sqrt{1 - \frac{v^2}{c^2}}$$

$$\Delta t_{B,Y} = \frac{l_0 \sqrt{1 - \frac{v^2}{c^2}}}{2(c + v)} = \frac{l_0}{2c} \sqrt{\frac{1 - \frac{v}{c}}{1 + \frac{v}{c}}}$$



One can see $\Delta t_{A,Y} > \Delta t_{B,Y}$

We can conclude the two events are simultaneous with respect to observer A but if these two events are not simultaneous with respect to observer B

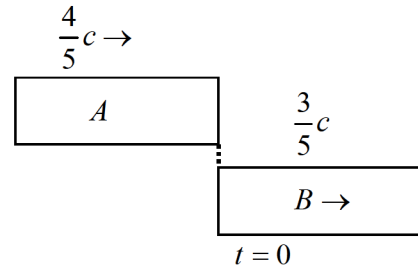
Example: Two trains A and B having same proper length l_0 are moving with respect to platform in parallel track with speed $\frac{4}{5}c$ and $\frac{3}{5}c$ respectively. At $t=0$ Engine of train A is just behind the last buggy of train B .

(a) If observer is attached to platform what will be time measured such that train A will completely crossed the train B

(b) If observer is attached to train B what will be time measured such that train A will completely crossed the train B .

(Train A will completely cross to train B means engine of train B will just behind the last buggy of train A)

Solution: (a) Speed of train A and train B with respect to observer is $\frac{4}{5}c$ and $\frac{3}{5}c$. Let's assume in time t train A will cross train B . Length of both trains are contracted with respect to observer.



Using kinematics relation $\frac{4}{5}c \cdot t = l_0 \sqrt{1 - \frac{16}{25}} + l_0 \sqrt{1 - \frac{9}{25}} + \frac{3}{5}ct$

$$\frac{c}{5}t = l_0 \times \frac{3}{5} + l_0 \times \frac{4}{5} \Rightarrow t = \frac{7l_0}{c}$$

(b) Velocity of train A with respect to platform $v_{A,P} = \frac{4}{5}c$

Velocity of train B with respect to platform $v_{B,P} = \frac{3}{5}c$

We need to find speed of A with respect to B $v_{A,B} = ?$

So we assume B is equivalent to S frame so platform will equivalent to S' frame

$v_{S',S} = v = -\frac{3}{5}c \hat{i}$, $v_{A,P} = v_{A,S'} = u'_x = \frac{4}{5}c$ and $v_{A,B} = v_{A,S} = u_x = ?$

Using formula $u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{\frac{4c}{5} - \frac{3c}{5}}{1 - \frac{1}{c^2} \times \frac{4c}{5} \times \frac{3c}{5}} = \frac{\frac{c}{5}}{\frac{13}{25}} = \frac{5}{13}c$

With respect to observer the length of A will contracted but length of B will not change.

Let us assume total time t such that train A will pass train B with respect to observer using kinematics relation

$$\frac{5}{13}c \times t = l_0 \sqrt{1 - \frac{25}{169}} + l_0 \Rightarrow \frac{5}{13}ct = \frac{12}{13}l_0 + l_0 \Rightarrow \frac{5}{13}ct = \frac{25}{13}l_0 \Rightarrow t = \frac{5l_0}{c}$$

Chapter Two

MASS ENERGY EQUIVALENCE

2.1 Derivation of Relativistic Mass

There are two identical particles A and B , where the rest mass of particle A is $m_{A,R} = m_0$ and rest mass of particle B is $m_{B,R} = m_0$.

There are two inertial frame S and S' where S' is moving with velocity with respect to S is $v\hat{i}$.

$-v_A\hat{j}$ is velocity of A from S frame. and $v'_B\hat{i}$ is velocity of B from S' frame. They are separated by distance Y at $t = 0$. If T_0 is time in which A and B travel distance Y from S and

S' frame respectively. So $v_A = v'_B = \frac{Y}{T_0}$

From symmetry we can conclude that initial momentum of system is zero..

From S' frame, $v_{B,S'} = v_B\hat{j}$ $m_{B,S'} = m'_B$

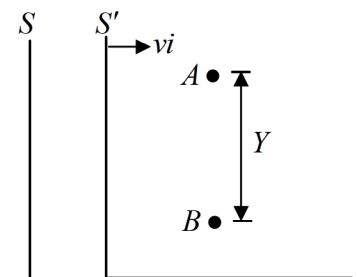


Figure 2.1

From S frame, $v_{S',S} = v\hat{i}$, $v_{A,S} = -v_A\hat{j}$, $m_{A,S} = m_A$

$$\vec{v}_{B,S} = v\hat{i} + v'_B\sqrt{1-\frac{v^2}{c^2}}\hat{j} = v\hat{i} + v_B\hat{j} \quad \text{where } v_B = v'_B\sqrt{1-\frac{v^2}{c^2}}$$

$$v_A = \frac{Y}{T_0} \quad v_B = \frac{Y}{T} = \frac{Y}{T_0}\sqrt{1-\frac{v^2}{c^2}}$$

From conservation of momentum

$$m_A v_A = m_B v_B = m_A \frac{Y}{T_0} = m_B \frac{Y}{T_0}\sqrt{1-\frac{v^2}{c^2}} \Rightarrow m_B = \frac{m_A}{\sqrt{1-\frac{v^2}{c^2}}}$$

Suppose A is rest with respect to A , $m_A = m_{A,R} = m_0 = m_{B,R}$

$$m_B = \frac{m_A}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{m_{B,R}}{\sqrt{1-\frac{v^2}{c^2}}} \Rightarrow m_B > m_{B,R} \quad \text{so } m(v) = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$$

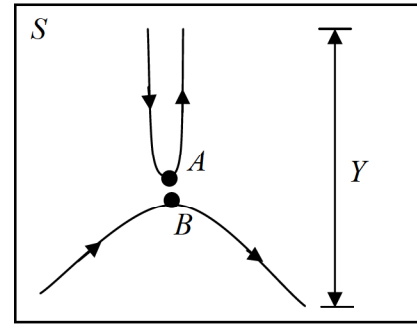


Figure 2.2

Relativistic Mass

The mass of a body moving at a speed v relative to an observer is larger than its mass when it is at rest relative to the observer.

$$m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}, \quad \text{where } m_0 \text{ is rest mass of body and } m \text{ is observed mass.}$$

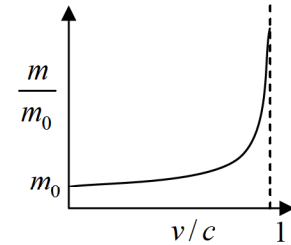


Figure 2.3

Relativistic Momentum

$$\vec{p} = m\vec{v} = \frac{m_0\vec{v}}{\sqrt{1-\frac{v^2}{c^2}}}$$

Relativistic force in one dimension and Newton's law of motion

The force is defined as $F = \frac{dp}{dt}$ where p is relativistic momentum.

$$F = \frac{dp}{dt} = \frac{d(mv)}{dt} = \frac{d}{dt} \left(\frac{m_0 v}{\sqrt{1-\frac{v^2}{c^2}}} \right) = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} \frac{dv}{dt} + m_0 v \frac{d}{dt} \left(1-\frac{v^2}{c^2} \right)^{-1/2}$$

$$\frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} \frac{dv}{dt} + \frac{m_0 v}{\left(1-\frac{v^2}{c^2} \right)^{3/2}} \frac{v}{c^2} \frac{dv}{dt} = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} \frac{dv}{dt} \left(1 + \frac{v^2}{1-\frac{v^2}{c^2}} \right) = \frac{m_0}{\left(1-\frac{v^2}{c^2} \right)^{3/2}} \frac{dv}{dt}$$

Relativistic Kinetic Energy

Using work energy theorem $T = \int F \cdot ds = \int_0^s \frac{dp}{dt} dx = \int_0^v dm \cdot u \frac{dx}{dt} = \int_0^v dm \cdot u$

$$T = \int_0^v u dm \cdot u + m \cdot u \cdot du$$

$$m = \frac{m_0}{\sqrt{1 - \frac{u^2}{c^2}}} = m^2 \left(1 - \frac{u^2}{c^2} \right) = m_0^2 \Rightarrow m^2 c^2 - m^2 u^2 = m_0^2 c^2 \Rightarrow 2mdmc^2 - 2mdmu^2 - m^2 2udu = 0$$

$$dmc^2 = dm u^2 + m u du = T = \int_0^v u dm \cdot u + m \cdot u \cdot du = \int_{m=m_0}^{m(v)} c^2 dm = m(v) c^2 - m_0 c^2 \Rightarrow T = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}} - m_0 c^2$$

Case when $\frac{v}{c} \rightarrow 0$,

$$T = m_0 c^2 \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right) = m_0 c^2 \left(\left(1 - \frac{v^2}{c^2} \right)^{-1/2} - 1 \right) = m_0 c^2 \left(1 + \frac{1}{2} \frac{v^2}{c^2} \dots - 1 \right) = \frac{1}{2} m_0 v^2$$

2.2 Rest Mass Energy and Relativistic Energy

Einstein suggested that, if particle is rest with respect to observer then he will measure rest mass m_0 and equivalent energy is measured as rest mass energy as $m_0 c^2$.

If particle of rest mass m_0 moving with observer then with speed v then equivalent energy is

identified as relativistic energy as $E = mc^2 \Rightarrow \frac{m_0 c^2}{\sqrt{1 - \left(\frac{v^2}{c^2} \right)}}$

Relationship between Total Energy and Momentum for Free Particle

Total energy and momentum are given by $E = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad p = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$

$$\Rightarrow p^2 \left(1 - \left(\frac{v^2}{c^2} \right) \right) = m_0^2 v^2 \Rightarrow m_0^2 v^2 + \frac{p^2 v^2}{c^2} = p^2 \Rightarrow v^2 = \frac{p^2 c^2}{m_0^2 c^2 + p^2} \Rightarrow v = \left(\frac{p^2 c^2}{m_0^2 c^2 + p^2} \right)^{1/2}$$

Putting the value of $v^2 = \frac{p^2 c^2}{m_0^2 c^2 + p^2}$ in equation $E = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$

We get the equation as $E^2 = m_0^2 c^4 + p^2 c^2 \Rightarrow E = \left(m_0^2 c^4 + p^2 c^2 \right)^{1/2}$

Examples of Mass Energy Equivalence

Example: A cube of density ρ_0 in rest frame moving with velocity v with respect to observer parallel to one of its edge. What is density measured by observer?

Solution: In rest frame $\frac{m_0}{V_0} = \rho_0$ and from moving frame $\rho = \frac{\text{mass}}{\text{volume}}$, where m is relativistic

mass given by, $m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$ and v is relativistic volume, $V = l_0 l_0 \cdot l_0 \sqrt{1 - \frac{v^2}{c^2}}$

$$\rho = \frac{\frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}}{V_0 \sqrt{1 - \frac{v^2}{c^2}}} = \frac{m_0}{V_0 \left(\sqrt{1 - \frac{v^2}{c^2}} \right)^2} \Rightarrow \rho = \frac{\rho_0}{1 - \frac{v^2}{c^2}} \quad \left[\because \frac{m_0}{V_0} = \rho_0 \right]$$

Example: Show that the rest mass of particle of momentum p and kinetic energy T is given by

$$m_0 = \frac{p^2 c^2 - T^2}{2Tc^2}$$

Solution: $E = E_K + E_0 \Rightarrow E = T + m_0 c^2$ (i)

$$E^2 = p^2 c^2 + m_0^2 c^4 \quad \text{(ii)}$$

Thus, from (i) and (ii), $m_0 = \frac{p^2 c^2 - T^2}{2Tc^2}$

Example: In the laboratory frame a particle P at rest mass m_0 is moving in the positive x -direction with speed $\frac{5c}{19}$. It approaches an identical particle Q which is moving in the negative x -direction with a speed of $\frac{2c}{5}$.

(a) What is speed of the particle P in the rest frame of the particle Q .

(b) What is Energy of the particle P in the rest frame of the particle Q .

Solution: (a) assume Q is S frame so laboratory can be treated as S' frame so

$$v_{L,Q} = v_{S',S} = v = -\left(-\frac{2}{5}c\right)\hat{i} \Rightarrow v = \frac{2}{5}c\hat{i},$$

Speed of P with respect to lab is $v_{P,L} = v_{P,S'} = u'_x = \frac{5}{19}c$

$$u'_x = \frac{5}{19}c, v = \frac{2}{5}c, u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{3}{5}c$$

$$(b) v_{P,Q} = u_x = \frac{3c}{5}, E = \frac{m_0 c^2}{\sqrt{1 - \frac{v_1^2}{c^2}}} = \frac{5}{4}m_0 c^2$$

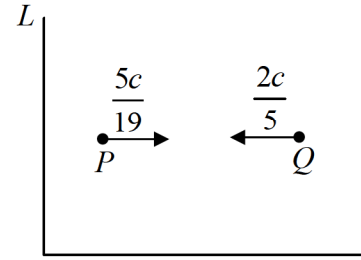


Figure 2.4

Example: The relativistic mass m of moving particle is $\frac{2m_0}{\sqrt{3}}$, where m_0 is its rest mass. Then what is linear momentum of particle?

Solution: and mass, $m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow \frac{2m_0}{\sqrt{3}} = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow v = \frac{c}{2}$

Momentum, $p = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{m_0 v}{\sqrt{1 - \frac{1}{4}}} = \frac{m_0}{\sqrt{1 - \frac{1}{4}}} \times \frac{1}{2}c \Rightarrow p = \frac{m_0 c}{\sqrt{3}}$

Example: A particle of rest mass m_0 moving with momentum $\frac{3m_0 c}{4}$.

(a) Find the total relativistic energy of particle.

(b) Find the speed of particle and verify momentum $p = \frac{3}{4}m_0 c$

(c) Find the kinetic energy of particle.

Solution: (a) Momentum is $p = \frac{3m_0c}{4}$ and rest mass energy is m_0c^2

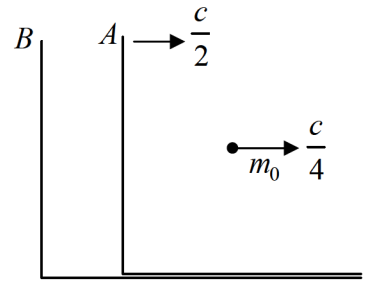
$$E = \sqrt{p^2c^2 + m_0^2c^4} = E = \sqrt{\frac{9}{16}m_0^2c^4 + m_0^2c^4} = \frac{5}{4}m_0c^2$$

$$(b) E = mc^2 \Rightarrow E = \frac{m_0c^2}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{5}{4}m_0c^2 = \frac{m_0c^2}{\sqrt{1-\frac{v^2}{c^2}}} \Rightarrow 1 - \frac{v^2}{c^2} = \frac{16}{25} \Rightarrow v = \frac{3}{5}c$$

$$p = mv = \frac{m_0v}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{m_0 \cdot \frac{3}{5}c}{\sqrt{1-\frac{9}{25}}} = \frac{3}{4}m_0c$$

$$(c) \text{ Relativistic kinetic energy } T = E - m_0c^2 = \frac{5}{4}m_0c^2 - m_0c^2 = \frac{m_0c^2}{4}$$

Example: A particle of rest mass m_0 moving with speed $\frac{c}{4}$ with respect to frame A in positive $\hat{x}$ direction. where frame B is moving with velocity $\frac{c}{2}$ with respect frame A in positive $\hat{x}$ direction.



(a) Find the relativistic energy of particle measured by observer attached with frame A

(b) Find the relativistic energy of particle measured by observer attached to frame B

Solution: (a) The rest mass of particle is m_0 . If observer is attached to frame A the speed of

mass with respect to mass is $\frac{c}{2}$, so relativistic mass $m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{m_0}{\sqrt{1-\frac{1}{16}}} = \sqrt{\frac{16}{15}}m_0$

so relativistic energy is $E = mc^2 = \sqrt{\frac{16}{15}}m_0c^2$

(b) It is given $v_{m,A} = \frac{c}{4}\hat{x}$, $v_{A,B} = \frac{c}{2}\hat{x}$ $v_{m,B} = ?$

We assume frame B is equivalent to S frame so then frame A is equivalent to S' frame.

$$v_{A,B} = v_{S',S} = v = \frac{c}{2}\hat{x}, v_{m,A} = v_{m,S'} = u'_x = \frac{c}{4}\hat{x}, v_{m,B} = v_{m,S} = u_x = ?$$

Using formula of relative speed $u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{\frac{c}{2} + \frac{c}{4}}{1 + \frac{c}{4} \cdot \frac{c}{2} \cdot \frac{1}{c^2}} = \frac{2c}{3}$

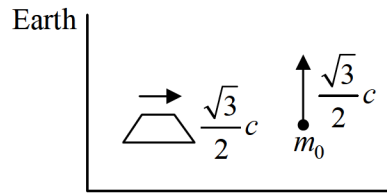
So, observer attached to frame B will measured that mass is moving the speed $\frac{2c}{3}$.

The relativistic mass measured by observer B is $m = \frac{m_0}{\sqrt{1 - \frac{|v_{m,A}|^2}{c^2}}} = \frac{m_0}{\sqrt{1 - \frac{4}{9}}} = \frac{3m_0}{\sqrt{5}}$

So relativistic energy is given by $E = mc^2 = \frac{3m_0 c^2}{\sqrt{5}}$

Example: A space ship is moving with respect to earth with speed $\frac{\sqrt{3}}{2}c$ in positive $\hat{x}$ direction.

A particle of rest mass m_0 moving with respect to earth speed $\frac{\sqrt{3}}{2}c$ in positive $\hat{y}$ direction.



(a) Find the mass of particle with respect to observer attached to earth.

(b) Find the mass of particle with respect to observer.

Solution: (a) If observer is attached to earth, he will measure that speed of particle is

$$v_{P,E} = \frac{\sqrt{3}}{2}c$$

so relativistic mass $m = \frac{m_0}{\sqrt{1 - \frac{|v_{P,E}|^2}{c^2}}} = \frac{m_0}{\sqrt{1 - \frac{3}{4}}} = 2m_0$

(b) If observer is attached to spaceship then we need to find speed of particle with respect to spaceship i.e. $v_{P,S} = ?$

It is given $v_{S,E} = \frac{\sqrt{3}}{2}c \hat{x}$ $v_{P,E} = \frac{\sqrt{3}}{2}c \hat{y}$

We assume space ship is S' frame, so earth will be treated as S $v_{S',S} = v = -\frac{\sqrt{3}}{2}c \hat{x}$

So $\vec{v}_{P,E} = \vec{v}_{P,S'} = u'_x = 0, u'_y = \frac{\sqrt{3}}{2}c, u'_z = 0, \vec{v}_{P,S} = ?$

$$u_x = \frac{u'_x + v}{1 + \frac{u'_x \cdot v}{c^2}} = -\frac{\sqrt{3}}{2}c, u_y = \frac{u'_y \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{u'_y \cdot v}{c^2}} = \frac{\sqrt{3}}{2}c \sqrt{1 - \frac{3}{4}} = \frac{\sqrt{3}}{4}c, u'_z = 0$$

$$\vec{v}_{P,S} = u_x \hat{x} + u_y \hat{y} + u_z \hat{z} = -\frac{\sqrt{3}}{2}c \hat{x} + \frac{\sqrt{3}}{4}c \hat{y}, |\vec{v}_{P,S}| = \sqrt{u_x^2 + u_y^2 + u_z^2} = \sqrt{\frac{3}{4} + \frac{3}{16}}c = \sqrt{\frac{15}{16}}c$$

$$m = \frac{m_0}{\sqrt{1 - \frac{|\vec{v}_{P,S}|^2}{c^2}}} = \frac{m_0}{\sqrt{1 - \frac{15}{16}}} = 4m_0$$

Example: if kinetic energy of particle is equal to rest mass energy of particle, then find the speed of particle.

Solution: Let us assume the rest mass energy of particle is $m_0 c^2$ and relativistic energy is

$$mc^2 = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Kinetic energy $T = mc^2 - m_0 c^2$

$$\Rightarrow m_0 c^2 = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}} - m_0 c^2 \Rightarrow 2 = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow 1 - \frac{v^2}{c^2} = \frac{1}{4} \Rightarrow v = \frac{\sqrt{3}}{2}c$$

Example: A π -meson at rest decays into μ meson mass and neutrino ν .

If rest mass of π meson, μ meson and neutrino are m_π, m_μ and m_ν then find the total relativistic energy of outgoing μ meson and neutrino.



Solution: It is given rest mass of π meson, μ meson and neutrino are m_π, m_μ and m_ν .

Let us assume E_μ and E_ν are energies of outgoing μ meson and neutrino

$$E_\pi = m_\pi c^2, E_\mu^2 = p_\mu^2 c^2 + m_\mu^2 c^4 \Rightarrow E_\mu^2 = p^2 c^2 + m_\mu^2 c^4$$

$$E_\nu^2 = p_\nu^2 c^2 + m_\nu^2 c^4 \Rightarrow E_\nu^2 = p^2 c^2 + m_\nu^2 c^4$$

From conservation of momentum $\vec{P}_\nu = -\vec{P}_\mu \Rightarrow |\vec{P}_\nu| = |\vec{P}_\mu| = P$

From conservation of energy $E_\pi = E_\mu + E_\nu \Rightarrow E_\mu + E_\nu = m_\pi c^2$

Using $E_\mu^2 - E_\nu^2 = (m_\mu^2 - m_\nu^2)c^4 \Rightarrow (E_\mu - E_\nu)(E_\mu + E_\nu) = (m_\mu^2 - m_\nu^2)c^4 \Rightarrow E_\mu - E_\nu = \frac{(m_\mu^2 - m_\nu^2)c^4}{m_\pi c^2}$

Solving equation $E_\mu - E_\nu = \frac{(m_\mu^2 - m_\nu^2)c^2}{m_\pi}$ and $E_\mu + E_\nu = m_\pi c^2$

$$E_\mu = \frac{1}{2m_\pi} [m_\pi^2 + m_\mu^2 - m_\nu^2] c^2 \quad E_\nu = \frac{1}{2m_\pi} [m_\pi^2 - m_\mu^2 + m_\nu^2] c^2$$

Example: A π -meson at rest decays in to μ meson mass and neutrino ν .

If rest mass of π meson, μ meson are m_π, m_μ and neutrino is mass less.

(a) Find the total relativistic energy of outgoing μ meson and neutrino.

(b) Find the magnitude of momentum of outgoing μ meson and neutrino.

(c) Find the speed of outgoing μ meson and neutrino.

Solution: (a) It is given rest mass of π meson, μ meson and are m_π, m_μ and neutrino is massless.

Let us assume E_μ and E_ν are energies of outgoing μ meson and neutrino

$$E_\pi = m_\pi c^2, \quad E_\mu^2 = p_\mu^2 c^2 + m_\mu^2 c^4 \Rightarrow E_\mu^2 = p^2 c^2 + m_\mu^2 c^4$$

$$E_\nu = p_\nu c \Rightarrow E_\nu = pc$$

From conservation of momentum $\vec{P}_\nu = -\vec{P}_\mu \Rightarrow |P_\nu| = |P_\mu| = P$

From conservation of energy $E_\pi = E_\mu + E_\nu \Rightarrow E_\mu + E_\nu = m_\pi c^2$

Using $E_\mu^2 - E_\nu^2 = m_\mu^2 c^4 \Rightarrow (E_\mu - E_\nu)(E_\mu + E_\nu) = m_\mu^2 c^4 \Rightarrow E_\mu - E_\nu = \frac{m_\mu^2 c^4}{m_\pi c^2}$

Solving equation $E_\mu - E_\nu = \frac{m_\mu^2 c^2}{m_\pi}$ and $E_\mu + E_\nu = m_\pi c^2$

$$E_\mu = \frac{1}{2m_\pi} [m_\pi^2 + m_\mu^2] c^2 \quad \text{and} \quad E_\nu = \frac{1}{2m_\pi} [m_\pi^2 - m_\mu^2] c^2$$

(b) $E_\nu = p_\nu c \Rightarrow E_\nu = pc$ and $|P_\nu| = |P_\mu| = P$

$$E_\nu = \frac{1}{2m_\pi} [m_\pi^2 - m_\mu^2] c^2 = p_\nu c = \frac{1}{2m_\pi} [m_\pi^2 - m_\mu^2] c^2 \Rightarrow p_\nu = p_\mu = p = \frac{1}{2m_\pi} [m_\pi^2 - m_\mu^2] c$$

(c) Hence the neutrino is massless so it will be moving with speed of light i.e. $v_\nu = c$

Now we will calculate the speed of μ meson we know

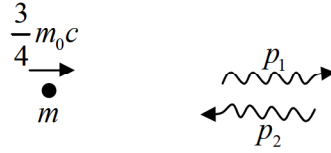
$$E_\mu = \frac{m_\mu c^2}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{(m_\pi^2 + m_\mu^2)c^2}{2m_\pi} \Rightarrow \sqrt{1 - \frac{v^2}{c^2}} = \frac{2m_\pi m_\mu}{(m_\pi^2 + m_\mu^2)}$$

$$\frac{v^2}{c^2} = 1 - \left(\frac{2m_\pi m_\mu}{m_\pi^2 + m_\mu^2} \right)^2 \Rightarrow \frac{(m_\pi^2 + m_\mu^2)^2 - 4m_\pi^2 m_\mu^2}{(m_\pi^2 + m_\mu^2)^2} \Rightarrow \frac{(m_\pi^2 - m_\mu^2)^2}{(m_\pi^2 + m_\mu^2)^2} = \left(\frac{m_\pi^2 - m_\mu^2}{m_\pi^2 + m_\mu^2} \right)^2$$

Example: A pion of rest mass m_0 moving with momentum $\frac{3}{4}m_0c$ completely decays into two photons. One photon will move in direction of pion and another is moving exactly opposite to pion.

(a) Find the momentum of outgoing photons.

(b) Find the energy of outgoing photons.



Solution: (a) The momentum of pion is $p = \frac{3}{4}m_0c$.

The energy of pion is $E_\pi = \sqrt{p^2 c^2 + m_0^2 c^4} = m_0 c^2 \left(\sqrt{\frac{9}{16} + 1} \right) = \frac{5}{4}m_0 c^2$

Let us assume p_1 and p_2 are momentum of outgoing photons.

The energy of outgoing photon $E_1 = p_1 c$ and $E_2 = p_2 c$.

From conservation of momentum $p_1 - p_2 = \frac{3}{4}m_0c$.

From conservation of energy $E_1 + E_2 = p_1 c + p_2 c = E_\pi = p_1 c + p_2 c = \frac{5}{4}m_0 c^2$

Solving equation $p_1 - p_2 = \frac{3}{4}m_0c$ and $p_1 + p_2 = \frac{5}{4}m_0c$

We get $p_1 = m_0c$ and $p_2 = \frac{m_0c}{4}$

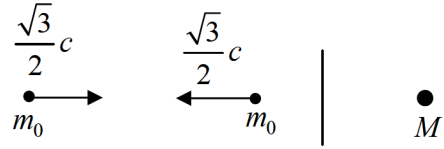
(b) The energy of outgoing photons are $E_1 = p_1 c = m_0 c^2$ and $E_2 = p_2 c = \frac{m_0 c^2}{4}$

Example: Two particles of same rest mass m_0 moving with

same speed with $\frac{\sqrt{3}}{2}c$ in opposite direction. They collide

head on and stick together and make composite lump.

Find the mass of composite lump.



Solution: Hence total momentum before Collision is zero then from conservation of momentum composite lump will remain rest with respect to observer. Rest mass of composite lump is M .

From conservation of energy $\frac{m_0 c^2}{\sqrt{1-\frac{v^2}{c^2}}} + \frac{m_0 c^2}{\sqrt{1-\frac{v^2}{c^2}}} = M c^2$ where $v = \frac{\sqrt{3}}{2}c$

$$\frac{m_0 c^2}{\sqrt{1-\frac{3}{4}}} + \frac{m_0 c^2}{\sqrt{1-\frac{3}{4}}} = M c^2 \Rightarrow M c^2 = 2m_0 c^2 + 2m_0 c^2 \Rightarrow M = 4m_0$$

Example: A heavy particle of rest mass M while moving along the positive z - direction, decays into two identical light particles with rest mass m (where $M > 2m$). Find the maximum value of the momentum that any one of the lighter particles can have in a direction perpendicular to the z - direction.

Solution: Let P be the momentum of heavy mass M , P_1 be the momentum of the light particles of mass m in the direction perpendicular to z and P_2 be the momentum in z -direction. According to conservation of momentum,

Momentum of mass M , $P = P_1 + P_2 = 2P_2 \Rightarrow P_2 = P/2$, Energy of mass M , $E = \sqrt{P^2 c^2 + M^2 c^4}$

Momentum of a mass $m = \sqrt{P_1^2 + P_2^2} = \sqrt{P_1^2 + \frac{P^2}{4}}$

Energy of mass m , $E_1 = \left(P_1^2 + \frac{P^2}{4} \right) c^2 + m^2 c^4$

As energy is conserved $E = E_1 + E_2 = 2E_1 \Rightarrow E_1 = \frac{E}{2} \quad \because E_1 = E_2$

Thus $E_1 = \frac{E}{2} = \left(P_1^2 + \frac{P^2}{4} \right) c^2 + m^2 c^4 \Rightarrow 4 \left(P_1^2 + \frac{P^2}{4} \right) c^2 + 4m^2 c^4 = P^2 c^2 + M^2 c^4$

$4P_1^2 c^2 + P^2 c^2 + 4m^2 c^4 = P^2 c^2 + M^2 c^4 \Rightarrow 4P_1^2 c^2 + 4m^2 c^4 = M^2 c^4 \Rightarrow 4P_1^2 = M^2 c^2 - 4m^2 c^2$

$P_1^2 = \frac{c^2}{4} (M^2 - 4m^2) \Rightarrow P_1 = \frac{c}{2} \sqrt{M^2 - 4m^2}$

Chapter Three

Relativistic Four Vectors

3.1 Space and Time

In special theory of relativity now we can consider the position vector is four-dimensional vector which can be given by 3 spatial (x, y, z) coordinate fourth -time coordinate (ict) .

From S frame the distance is identified as $s^2 = x^2 + y^2 + z^2 + (ict)^2 = x^2 + y^2 + z^2 - c^2 t^2$

Then it can be shown that $x^2 + y^2 + z^2 - c^2 t^2$ is invariant under Lorentz transformation

$$\begin{aligned} x^2 + y^2 + z^2 - c^2 t^2 &= \gamma^2 (x' + vt')^2 + (y')^2 + (z')^2 - c^2 \left(t' + v \frac{x'}{c^2} \right)^2 \gamma^2 \\ &= \gamma^2 \left(x'^2 + v^2 t'^2 + 2x'vt' - c^2 t'^2 - \frac{v^2 x'^2}{c^2} - 2x'vt' \right) + y'^2 + z'^2 \\ &= \frac{1}{\left(1 - \frac{v^2}{c^2} \right)} \left(x'^2 \left(1 - \frac{v^2}{c^2} \right) - c^2 t'^2 \left(1 - \frac{v^2}{c^2} \right) \right) + y'^2 + z'^2 = x'^2 + y'^2 + z'^2 - c^2 t'^2 \end{aligned}$$

$$x^2 + y^2 + z^2 - c^2 t^2 = x'^2 + y'^2 + z'^2 - c^2 t'^2 \text{ because } y = y', z = z'$$

$\Rightarrow x^2 - c^2 t^2 = x'^2 - c^2 t'^2$ which will analogically rotation of coordinate system.

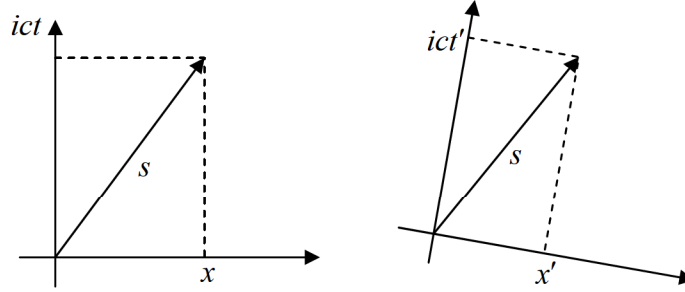


Figure 3.1

Space time interval $(\Delta s)^2 = c^2(\Delta t)^2 - (\Delta x)^2 = c^2(\Delta t')^2 - (\Delta x')^2 = c^2\tau_0^2$, where τ_0 is proper time interval.

(i) Space-like Intervals: Time separation between the two event is less than the time taken by light in covering the distance between them.

$$\frac{\Delta x}{c} > \Delta t$$

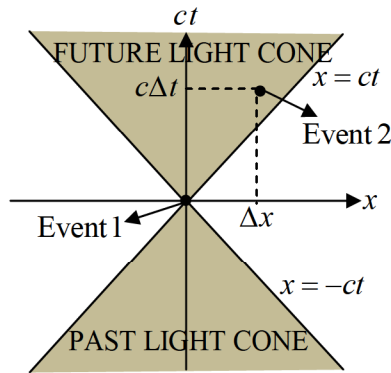


Figure 3.2

(ii) Time-like intervals: Time separation between two event is more than the time taken by light in covering the distance between them

$$\frac{\Delta x}{c} < \Delta t$$

(iii) Light-like intervals: Time separation between two event is equal to time taken by light in covering the distance between them

$$\frac{\Delta x}{c} = \Delta t$$

Chapter Three

Relativistic Four Vectors

3.2 Minkowski Diagrams

Minkowski Diagrams (sometimes called “spacetime” diagrams) are extremely useful in seeing how coordinates transform between different reference frames.

Let frame S' move at speed v with respect to frame S (along the x - axis, as usual, and ignore the y and z components). Draw the x and ct axes of frame S . What do the x' and ct' axes of S' look like, superimposed on this diagram?

That is, at what angles are the axes inclined, and what is the size of one unit on these axes? (There is no reason why one unit on the x' and ct' axes should have the same length on the paper as one unit on the x and ct axes). We can answer these questions by using the Lorentz transformations, We'll first look at the ct' axis, and then the x' axis.

ct' - axis angle and unit size

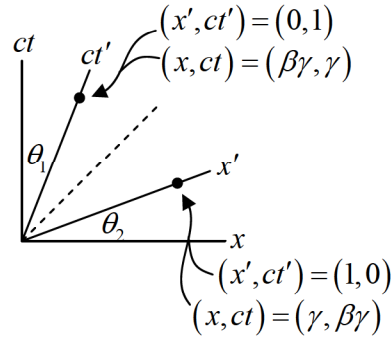
Look at the point $(x', ct') = (0, 1)$,

$$\text{where } \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \text{ and } \beta = \frac{v}{c}$$

$$\text{Equation } x = \frac{x' + vt'}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{0 + \frac{v}{c} \cdot 1}{\sqrt{1 - \frac{v^2}{c^2}}} = \beta\gamma$$

$$\text{and } t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow ct = \gamma$$

$$(x', ct') = (0, 1) \Rightarrow (x, ct) = (\gamma\beta, \gamma)$$



The angle between the ct' and ct axes is therefore given by $\tan \theta_1 = \frac{\gamma\beta}{\gamma} = \beta$. With $\beta = \frac{v}{c}$, we

have $\tan \theta_1 = \beta$

Look at the point $(x', ct') = (1, 0)$, Equation $x = \frac{x' + vt'}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1 + 0}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma$ and

$$t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{0 + \frac{v}{c} \cdot 1}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow ct = \frac{\frac{v}{c}}{\sqrt{1 - \frac{v^2}{c^2}}} = \beta\gamma$$

$$(x', ct') = (1, 0) \Rightarrow (x, ct) = (\gamma, \gamma\beta)$$

If θ_2 is angle between x and x' then $\tan \theta_2 = \frac{\beta\gamma}{\gamma} = \beta$

Alternatively,

The ct' axis is simply the “worldline” of the origin of S' (A worldline is the path an object takes as it travels through spacetime). The origin moves at speed v with respect to S . Therefore, points on the ct' axis satisfy $x/t = v$, or $x/ct = v/c$.

On the paper, the point $(x', ct') = (0, 1)$, which we just found to be the point $(x, ct) = (\gamma\beta, \gamma)$, is a distance $\gamma\sqrt{1 + v^2/c^2}$ from the origin.

Therefore, using the definitions of β and γ , we see that $\frac{\text{one } ct' \text{ unit}}{\text{one } ct \text{ unit}} = \sqrt{\frac{1 + \beta^2}{1 - \beta^2}}$

as measured on a grid where the x axes are orthogonal. This ratio approaches infinity as $\beta \rightarrow 1$.

And it of course equals 1 if $\beta = 0$.

x' - axis angle and unit size

The same basic argument holds here. Look at the point $(x', ct') = (1, 0)$, which lies on the

x' - axis, one x' unit from the origin. Equations $x = \frac{x' + vt'}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma, t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{\gamma v}{c^2} \Rightarrow ct = \frac{\gamma v}{c}$

tell us that this point is the point $(x, ct) = (\gamma, \gamma\beta)$. The angle between the x' and x axes is therefore given by $\tan \theta_2 = ct / x = v / c$. So as in the ct' - axis case, $\tan \theta_2 = \beta$

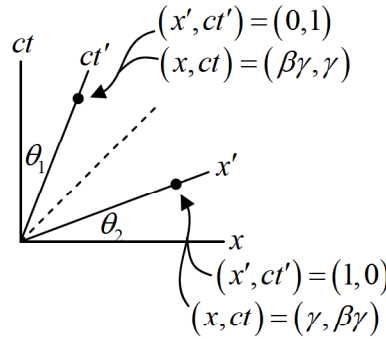


Figure 3.4

On the paper, the point $(x', ct') = (1, 0)$, which we just found to be the point $(x, ct) = (\gamma, \gamma v / c)$, is a distance $\gamma \sqrt{1 + v^2 / c^2}$ from the origin. So, as in the ct' - axis case,

$$\frac{\text{one } x' \text{ unit}}{\text{one } x \text{ unit}} = \sqrt{\frac{1 + \beta^2}{1 - \beta^2}}$$

as measured on a grid where the x and ct axes are orthogonal. Both the x' and ct' axes are therefore stretched by the same factor, and tilted in by the same angle, relative to the x and ct axes. This “squeezing in” of the axes in a Lorentz transformation is different from what happens in a rotation, where both axes rotate in the same direction.

Chapter Three

Relativistic Four Vectors

3.3 Symmetry in Momentum and Energy

We have seen that $c^2 t^2 - r^2 = c^2 \tau_0^2$ is invariant under Lorentz transformation similarly

We have relation $E^2 = p^2 c^2 + m_0^2 c^4 \Rightarrow E^2 - p^2 c^2 = m_0^2 c^4$ which means rest mass m_0 is also invariant under Lorentz transformation

On the basis of symmetry, we can also discuss write Lorentz transformation for momentum vector $\vec{p}$ and energy E .

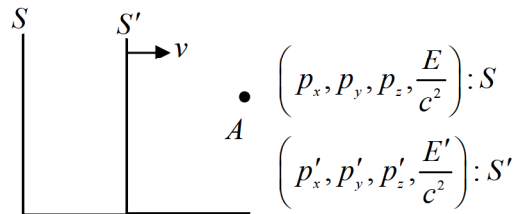


Figure 3.5

If momentum p_x, p_y, p_z and energy E are measured from S frame. p'_x, p'_y, p'_z and E' are momentum and energy measured from S' frame

Then there are equivalent Lorentz transformation for energy and momentum as

$\left(p_x, p_y, p_z, \frac{E}{c^2}\right)$ and $\left(p'_x, p'_y, p'_z, \frac{E'}{c^2}\right)$ which is given as

$$p_x = \frac{p'_x + \frac{vE'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad p_y = p'_y \quad p_z = p'_z \quad \frac{E}{c^2} = \frac{\frac{E'}{c^2} + \frac{vp'_x}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow E = \frac{E' + vp'_x}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$p'_x = \frac{p_x - \frac{vE}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad p'_y = p_y \quad p'_z = p_z \quad \frac{E'}{c^2} = \frac{\frac{E}{c^2} - \frac{vp_x}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow E' = \frac{E - vp_x}{\sqrt{1 - \frac{v^2}{c^2}}}$$

3.4 Doppler Effect in Light

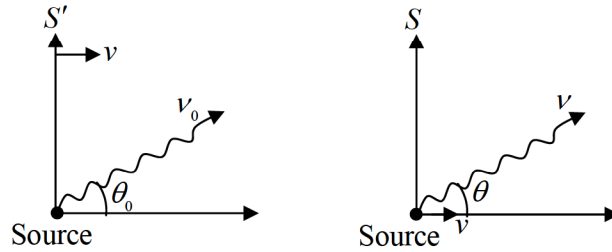


Figure 3.6

ν_0 is frequency measured by observer when there is not any relative speed between source and observer

v is the relative speed between source of light and observer

θ is the angle between velocity of light ($\vec{c} = c \cos \theta \hat{i} + c \sin \theta \hat{j}$) and relative velocity $\vec{v} = v \hat{i}$ between source and observer

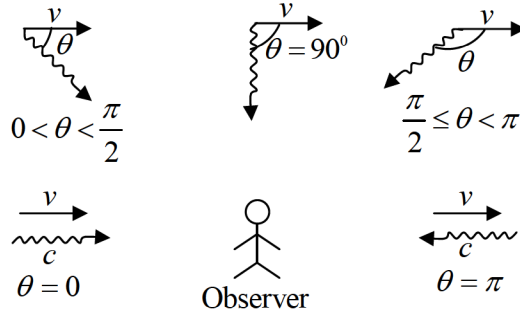
From S' frame i.e., when observer is attached to rest frame of source then $E = h\nu_0$

From S frame i.e., when observer and source has relative velocity

$$E = h\nu, \quad p_x = \frac{E}{c} \cos \theta = \frac{h\nu}{c} \cos \theta,$$

Using Lorentz transformation between energy and momentum

$$E' = \frac{E - p_x v}{\sqrt{1 - \frac{v^2}{c^2}}} = h\nu_0 = \frac{h\nu - \frac{h\nu \cos \theta}{c} v}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow \nu = \frac{\nu_0 \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c} \cos \theta}$$



3.5 Longitudinal Doppler's effect

Case 1: $\theta = 0$ $\nu = \nu_0 \frac{\sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c}} = \nu_0 \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}}$ for wave length $\lambda = \lambda_0 \sqrt{\frac{1 - \frac{v}{c}}{1 + \frac{v}{c}}}$

Case 2: $\theta = \pi$ $\nu = \nu_0 \frac{\sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{v}{c}} = \nu_0 \sqrt{\frac{1 - \frac{v}{c}}{1 + \frac{v}{c}}}$ for $\lambda = \lambda_0 \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}}$

Transverse Doppler's effect

Case 3: $\theta = \frac{\pi}{2}$ $\nu = \nu_0 \sqrt{1 - \frac{v^2}{c^2}}$ for $\lambda = \frac{\lambda_0}{\sqrt{1 - \frac{v^2}{c^2}}}$

Example: Consider a radioactive nucleus that is travelling at a speed $\frac{c}{2}$ with respect to the lab frame. It emits γ - rays of frequency ν_0 in its rest frame. There are two stationary detectors, (first detector A is not on the path of the nucleus in the lab .the second detector B is in path of nucleus such that nucleus approach it).

- (a) If a γ - ray photon is emitted when the nucleus is closest to the detector, its observed frequency at the first detector is ν_1 , find the value of ν_1
- (b) The second detector observed frequency is ν_2 find ν_2
- (c) What will measurement of frequency by detector B if nucleus is moving away from detector B ?

Solution: (a) If detector A is not in path then closet distance is when nuclear is perpendicular to detector A so it is case of transverse Doppler's effect, so

$$\nu_1 = \nu_0 \sqrt{1 - \frac{v^2}{c^2}} \quad \nu_1 = \nu_0 \sqrt{1 - \frac{1}{4}} \Rightarrow \nu_1 = \nu_0 \frac{\sqrt{3}}{2}$$

(b) Second detector B is in path of nucleus so there is longitudinal Doppler effect .The nuclear is

moving towards the detector B so $\nu_2 = \nu_0 \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}} = \nu_0 \sqrt{\frac{1 + \frac{1}{2}}{1 - \frac{1}{2}}} = \sqrt{3}\nu_0$

(c) Second detector B is in path of nucleus so there is longitudinal Doppler effect .The nuclear is

moving away the detector B so $\nu_2 = \nu_0 \sqrt{\frac{1 - \frac{v}{c}}{1 + \frac{v}{c}}} = \nu_0 \sqrt{\frac{1 - \frac{1}{2}}{1 + \frac{1}{2}}} = \frac{\nu_0}{\sqrt{3}}$

Example: A distant galaxy in constellation Hydra is receding from the earth at $6.12 \times 10^7 \text{ m/s}$ by how much is a green spectral line of wavelength 500 nm emitted by this galaxy shifted towards the red spectrum.

Solution: $\lambda = \lambda_0 \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}} \quad v = 0.204c \quad \lambda_0 = 500 \text{ nm}$

$\lambda = 500 \sqrt{\frac{1 + .204}{1 - .204}} = 615 \text{ nm}$ which is orange part of spectrum. The shift is $\lambda - \lambda_0 = 115 \text{ nm}$.

Properties of Four Vectors

(i) **Linear Combinations:** If A and B are 4-vectors, then $C = aA + bB$ is also a 4-vector

(ii) **Inner Product invariance:** Consider two arbitrary 4-vectors A and B . Define their inner product to be $A \cdot B = A_0 B_0 - A_1 B_1 - A_2 B_2 - A_3 B_3 = A_0 B_0 - \vec{A} \cdot \vec{B} = A' \cdot B'$, where A and B are measured from S frame and A' and B' are measured from S' frame.

(iii) **Norms:** For 4-vector A the norms is defined as

$$A^2 = A \cdot A = A_0 A_0 - A_1 A_1 - A_2 A_2 - A_3 A_3 = A_0^2 - |\vec{A}|^2$$

Four Vectors and Relativistic Invariance

- Four position vectors $ds = (dx, dy, dz, icdt)$
- Four velocity vectors $\frac{ds}{dt} = \left(\frac{dx}{d\tau}, \frac{dy}{d\tau}, \frac{dz}{d\tau}, \frac{icdt}{d\tau} \right) u = \gamma \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}, \frac{icdt}{dt} \right)$
 $u = \gamma(u_x, u_y, u_z, ic) \Rightarrow u = \gamma(\vec{u}, ic)$
- Four momentum, Four Energy vector: $P = \gamma(m_0 u, im_0 c)$
 $= (\vec{mu}, imc) = \left(\vec{p}, \frac{imc^2}{c} \right) \Rightarrow \vec{P} = \left(\vec{p}, \frac{iE}{c} \right)$
- Four Force: $F = \gamma \left(\vec{F}, \frac{idmc}{dt} \right)$
- Four dimensional space time continuum:

The square of interval is represented as

$$S_{12}^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 - c^2(t_2 - t_1)^2 = r_{12}^2 - c^2(t_2 - t_1)^2$$

Collisions and Decays

One can use concept of 4-vector as conservation of energy and momentum in decay or collision problem.

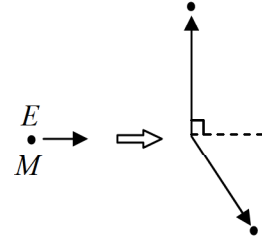
4-vector momentum $\vec{P}$ can be defined as $\vec{P} = \left(\frac{E}{c}, \vec{p} \right) = \left(\frac{E}{c}, p_x, p_y, p_z \right)$,

where $\left(\frac{E}{c} \right)^2 - |\vec{p}|^2 = m^2 c^2 \Rightarrow \vec{P} \cdot \vec{P} = m^2 c^2$.

Conservation of energy and momentum in a collision or decay problem reduce to concise

statement $\vec{P}_{before} = \vec{P}_{after}$

Example: A particle with mass M and energy E decays into two identical particles. In the lab frame, one of them is emitted at a 90° angle, as shown in figure. What are the energies of the created particles? (Take $c=1$)



Solution: The 4-momentum before the decay is

$$P = (E, p, 0, 0)$$

where $p = \sqrt{E^2 - M^2}$. Let the created particles have mass m , and let the second particle make an angle θ with the x -axis. The 4-momenta after the decay are

$$P_1 = (E_1, 0, p_1, 0), P_2 = (E_2, p_2 \cos \theta, -p_2 \sin \theta, 0)$$

With using the 4-momenta, conservation of energy and momentum can be combined into the statement, $\vec{P} = \vec{P}_1 + \vec{P}_2$. Therefore,

$$\begin{aligned} P - P_1 &= P_2 \Rightarrow (P - P_1) \cdot (P - P_1) = P_2 \cdot P_2 \Rightarrow P^2 - 2P \cdot P_1 + P_1^2 = P_2^2 \\ \Rightarrow M^2 - 2EE_1 + m^2 &= m^2 \Rightarrow E_1 = \frac{M^2}{2E} \end{aligned}$$

And then $E_2 = E - E_1 = (2E^2 - M^2)/2E$. This solution should convince you that 4-momenta can save you a lot of work. What happened here was that the expression for P_2 was fairly messy, but we arranged things so that it appeared only in the form of P_2^2 , which is simply m^2 . 4-momenta provide a remarkably organized method for sweeping unwanted garbage under the rug.

Chapter Four

Relativistic

Electrodynamics

4.1 Four Current Vectors

Relativistic Electromagnetism is a physical phenomenon explained in electromagnetic field theory due to coulomb's law, Maxwell four equation and Lorentz transformation.

The requirement that the equations remain consistent when viewed from various moving observers led to special relativity, a geometric theory of 4-space where intermeditation is by light and radiation.

Analogy

We have relation in space ($\vec{r}$) time (t) as $c^2 t^2 - r^2 = c^2 \tau_0^2$ where τ_0 is proper time

Similar relation we have for relativistic mass (m) and momentum (p) as $c^2 m^2 - p^2 = c^2 m_0^2$, where m_0 is rest mass.

On the basis of these analogy we have relation for charge density (ρ) and charge density $\vec{J}$ as $c^2 \rho^2 - j^2 = c^2 \rho_0^2$ where ρ_0 is identify as proper charge density which is measured from rest frame of charge .

$$\text{So } \rho = \frac{\rho_0 m}{m_0} \text{ and } \vec{j} = \frac{\rho_0}{m_0} \vec{p}$$

So, we have equivalent Lorentz transformation as well as inverse Lorentz transformation for current density $\vec{j}$ and charge density ρ .

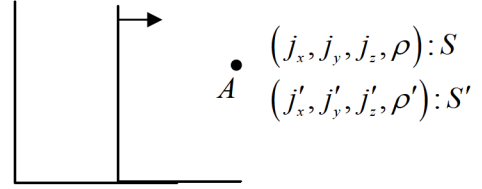


Figure 4.1

$$j'_x = \frac{j_x - \rho v}{\sqrt{1 - v^2/c^2}}, \quad j'_y = j_y, \quad j'_z = j_z$$

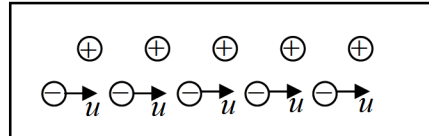
and $\rho = \frac{\rho' - v j'_x / c^2}{\sqrt{1 - v^2/c^2}}$

Similarly, we have inverse Lorentz transformation

$$j_x = \frac{j'_x + \rho' v}{\sqrt{1 - v^2/c^2}}, \quad j_y = j'_y, \quad j_z = j'_z \quad \text{and} \quad \rho = \frac{\rho' + v j'_x / c^2}{\sqrt{1 - v^2/c^2}}$$

Example: Consider a long straight wire rest in S frame. The free electrons moves with drift velocity u towards right as shown in figure. If number of free electron per unit volume is n .

- (a) Find the net charge density from S frame
- (b) Find net current density from S
- (c) If another inertial frame S' moving with respect to S with velocity v as same direction of Drift velocity u Find total charge density with respect to S'
- (d) If another inertial frame S' moving with respect to S with velocity v as same direction of Drift velocity u Find total charge density with respect to S'



Solution: From S frame negative charge density is due to free electron is $\rho^- = \text{negative}$ there is similar number of positive charge $\rho^+ = +ne$ the positive charge is rest with respect to S frame. The distance between negative electron and positive charge is same.

(a) $\rho = \rho^- + \rho^+ = (-ne) + (ne) = 0$

(b) $j_x^- = \rho^- u = -neu$, $j_x^+ = 0$

(c) $\rho'^- = \frac{\rho^- - v j_x^- / c^2}{\sqrt{1 - v^2/c^2}} = \frac{\rho^- - v \rho^- u / c^2}{\sqrt{1 - v^2/c^2}} \Rightarrow \rho'^- = \rho^- \frac{(1 - vu/c^2)}{\sqrt{1 - v^2/c^2}}$ where $j_x^- = \rho^- u$

$\rho'^+ = \frac{\rho^+ - v j_x^+ / c^2}{\sqrt{1 - v^2/c^2}} = \frac{\rho^+}{\sqrt{1 - v^2/c^2}}$ where

$$\rho' = \rho'^+ + \rho'^- \Rightarrow \rho' = \frac{ne}{\sqrt{1 - v^2/c^2}} - \frac{ne(1 - vu/c^2)}{\sqrt{1 - v^2/c^2}} = \frac{nevu/c^2}{\sqrt{1 - v^2/c^2}}$$

$$(d) \quad j_x^- = \frac{j_x^- - \rho^- v}{\sqrt{1 - v^2/c^2}}, = \frac{\rho^- u - \rho^- v}{\sqrt{1 - v^2/c^2}} = \rho^- \frac{u - v}{\sqrt{1 - v^2/c^2}} \Rightarrow j_x'^+ = -ne \frac{u - v}{\sqrt{1 - v^2/c^2}}$$

$$j_x'^+ = \frac{j_x^+ - \rho^+ v}{\sqrt{1 - v^2/c^2}}, = \frac{0 - \rho^+ v}{\sqrt{1 - v^2/c^2}} = \frac{-\rho^+ v}{\sqrt{1 - v^2/c^2}} = \frac{-nev}{\sqrt{1 - v^2/c^2}}$$

$$j_x' = j_x'^+ + j_x'^- = \frac{-nev}{\sqrt{1 - v^2/c^2}} + -ne \frac{u - v}{\sqrt{1 - v^2/c^2}} = \frac{-nue}{\sqrt{1 - v^2/c^2}}$$

Chapter Four

Relativistic

Electrodynamics

4.2 Transformation of electric and magnetic field

In one inertial frame S the charge density and current density of system measured. If the same system is measured from another inertial frame S' the charge and current density of same system is different. This transformation is according to Lorentz transformation. Hence charge density and current density are transformed it is very obvious that electric field and magnetic field will also be transformed.

If component of electric field $\vec{E}':(E'_x, E'_y, E'_z)$ and component of magnetic field $\vec{B}':(B'_x, B'_y, B'_z)$ are measured from S' frame then these component of electric field $\vec{E}:(E_x, E_y, E_z)$ and component of magnetic field $\vec{B}:(B_x, B_y, B_z)$ are measured from S .

The electric field is transformed as $E'_\parallel = E_\parallel$ $E'_\perp = \gamma[E_\perp + (v \times B)_\perp]$

$$\begin{aligned} E'_x &= E_x & E_x &= E'_x \\ E'_y &= \gamma(E_y - vB_z) & E_y &= \gamma(E'_y + vB'_z) \\ E'_z &= \gamma(E_z - vB_y) & E_z &= \gamma(E'_z + vB'_y) \end{aligned}$$

$$B'_{\parallel} = B_{\parallel} \quad \text{and} \quad B'_{\perp} = \gamma \left[B_{\perp} - \frac{1}{c^2} (v \times E)_{\perp} \right]$$

$$B'_x = B_x \qquad B'_y = B_y$$

$$B'_y = \gamma \left(B_y + \frac{v}{c^2} E_z \right), \quad B'_z = \gamma \left(B_z - \frac{v}{c^2} E_y \right)$$

$$B'_z = \gamma \left(B_z - \frac{v}{c^2} E_y \right), \quad B'_x = \gamma \left(B_x + \frac{v}{c^2} E_z \right)$$

Example: A test charge is rest in frame S outside a long straight stationary wire carrying a current. . The density of free electrons is n and drift velocity of electrons in wire is u

- (a) Find the net current and net charge in the wire from S frame
- (b) Find the electric field and magnetic field measured by observer from the rest frame of S at distance r perpendicular distance
- (c) if another frame S' moving with respect to S along length of wire, find electric field and magnetic field measured from rest frame of S'

Solution: (a) With respect to rest frame of S positive ions are rest but linear charge density of electron λ^- and linear charge density of positive ions λ^+ are same so there is no any net charge density. So, total charge density $\lambda = 0$. The current in rest frame of S frame is $i = \lambda^- u = -neu$

(b) With respect to rest frame of wire or from S frame, the total charge density is $\lambda = 0$, so electric field is zero i.e., $E_x = 0, E_y = 0, E_z = 0$

and magnetic field is $B_x = 0, B_y = 0, B_z = \frac{\mu_0 i}{2\pi r} = \frac{\mu_0 neu}{2\pi r}$

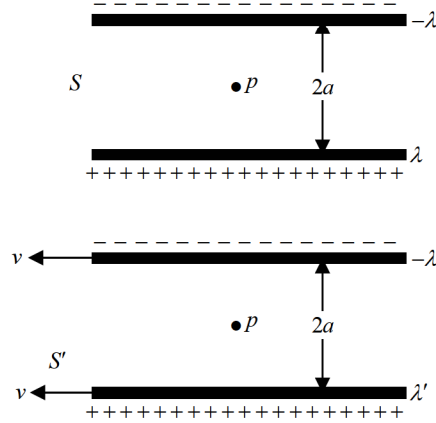
$$(c) \quad E'_x = E_x \quad E'_y = \gamma \left(E_y - v B_z \right), \quad E'_z = \gamma \left(E_z - v B_y \right)$$

$$\text{so} \quad E'_x = 0, \quad E'_y = -\gamma v B_z = -\gamma \frac{\mu_0 neu}{2\pi r} \quad E'_z = 0$$

$$B'_x = B_x \quad B'_y = \gamma \left(B_y + \frac{v}{c^2} E_z \right) \quad B'_z = \gamma \left(B_z - \frac{v}{c^2} E_y \right)$$

$$B'_x = B_x = 0 \quad B'_y = 0 \quad B'_z = \gamma \frac{\mu_0 neu}{2\pi r}$$

Example: consider two long parallel wires separated by a distance $2a$ bearing equal and opposite uniform charge distributions. In frame S at rest with respect to wires there is no current flowing and the linear charge density is λ



- Calculate the electric and magnetic fields in frame S at point P mid away between wires
- Frame S' moves at velocity parallel to length of wires (x -axis). Find the linear charge density λ' and current in the wires as measured in frame S' .
- Calculate the electric and magnetic fields measured in frame S' at mid away between wires.
- Verify the result from transformation of electric field and magnetic field.

Solution: (a) $E_x = 0, E_y = \frac{-\lambda}{2\pi\epsilon_0 a}(-\hat{y}) + \frac{\lambda}{2\pi\epsilon_0 a}\hat{y} = \frac{\lambda}{\pi\epsilon_0 a}\hat{y} \quad E_z = 0$

(b) $\lambda' = \frac{\lambda - \frac{vj_x}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{\lambda}{\sqrt{1 - \frac{v^2}{c^2}}}$ because $j_x = 0$

$j'_x = \frac{j_x - \lambda v}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{-\lambda v}{\sqrt{1 - \frac{v^2}{c^2}}}$

(c) $E'_x = 0 \quad E'_y = \frac{-\lambda'}{2\pi\epsilon_0 a}(-\hat{y}) + \frac{\lambda'}{2\pi\epsilon_0 a}\hat{y} = \frac{\lambda'}{\pi\epsilon_0 a}\hat{y} \Rightarrow E'_y = \frac{\lambda}{\pi\epsilon_0 a \sqrt{1 - \frac{v^2}{c^2}}}\hat{y}$

$B'_x = 0, B'_y = 0 \quad B'_z = \frac{\mu_0 j'_x}{2\pi a}\hat{z} + \frac{\mu_0 (-i)}{2\pi a} - \hat{z} = -\frac{\mu_0 \lambda v}{\pi a \sqrt{1 - \frac{v^2}{c^2}}}\hat{z}$

(d) $E'_x = E_x = 0, \quad E'_z = \gamma(E_z - vB_y) = 0$

$$E'_y = \gamma(E_y - vB_z) = \gamma \frac{\lambda}{\pi \epsilon_0 a} \Rightarrow E'_y = \frac{\lambda}{\pi \epsilon_0 a \sqrt{1 - \frac{v^2}{c^2}}},$$

$$B'_x = B_x = 0, \quad B'_y = \gamma\left(B_y + \frac{v}{c^2} E_z\right) \Rightarrow B'_y = 0$$

$$B'_z = \gamma\left(B_z - \frac{v}{c^2} E_y\right) \Rightarrow B'_z = -\gamma \frac{v}{c^2} \frac{\lambda}{\pi \epsilon_0 a} \Rightarrow B'_z = -\frac{\mu_0 \lambda v}{\pi a \sqrt{1 - \frac{v^2}{c^2}}} \text{ put } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

List of Lorentz Invariant Quantity:

- Proper time τ_0 is Lorentz invariant $c^2 \tau^2 - |\vec{r}|^2 = c^2 \tau_0^2$ where τ is time and $\vec{r}$ position vector
- Rest mass m_0 is Lorentz invariant $c^2 m^2 - |\vec{p}|^2 = c^2 m_0^2$ where m is relativistic mass and $\vec{p}$ is associated relativistic momentum vector
- Rest mass energy $m_0 c^2$ is Lorentz invariant $E^2 - |\vec{p}|^2 c^2 = m_0^2 c^4$ where E is total relativistic energy and $\vec{p}$ is associated momentum vector
- Rest charge ρ_0 is Lorentz invariant $c^2 \rho^2 - |\vec{j}|^2 = c^2 \rho_0^2$ where ρ is charge density and $\vec{j}$ is associated current density
- $|\vec{E}|^2 - c^2 |\vec{B}|^2$ and $\vec{E} \cdot \vec{B}$ are Lorentz invariant where $\vec{E}$ is electric field and $\vec{B}$ is magnetic field
- D'Alembert operator $\square^2 = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2$ is Lorentz invariant where ∇^2 is Laplacian operator

Example: An inertial observer A at rest measures the electric and magnetic field $E = (\alpha, 0, 0)$ and $B = (\alpha, 0, 2\alpha)$ in a region, where α is a constant. Another inertial observer B , moving with a constant velocity with respect to A , measures the fields as $E' = (E'_x, \alpha, 0)$ and $B' = (\alpha, B'_y, \alpha)$. Then in units $c = 1$, find E'_x and B'_y are given, respectively.

Solution: For given value of electric field $\vec{E}$ and $\vec{B}$ the quantity $\vec{E} \cdot \vec{B} |\vec{E}|^2 - |\vec{B}|^2$ is invariant

For inertial observer A electric field is $\vec{E} = \alpha \hat{i}$ and magnetic field is $\vec{B} = \alpha \hat{i} + 2\alpha \hat{k}$

$$\vec{E} \cdot \vec{B} = \alpha^2 \text{ and } |\vec{E}|^2 - |\vec{B}|^2 = \alpha^2 - \alpha^2 + 4\alpha^2 = |\vec{E}|^2 - |\vec{B}|^2 = 4\alpha^2$$

Now for inertial observer B ,

electric field is $\vec{E}' = E'_x \hat{i} + \alpha \hat{j}$ and magnetic field is $\vec{B}' = \alpha \hat{i} + B'_y \hat{j} + \alpha \hat{k}$

$$\text{So } \vec{E}' \cdot \vec{B}' = E'_x \alpha + B'_y \alpha \text{ and } |E'^2| - |B'^2| = (E'^2_x + \alpha^2) - (\alpha^2 + B'^2_y + \alpha^2) = E'^2_x - B'^2_y - \alpha^2$$

$$\vec{E} \cdot \vec{B} = \vec{E}' \cdot \vec{B}' \Rightarrow \alpha^2 = E'_x \alpha + B'_y \alpha \Rightarrow E'_x + B'_y = \alpha$$

$$|\vec{E}|^2 - |\vec{B}|^2 = |E'^2| - |B'^2| \Rightarrow 4\alpha^2 = E'^2_x - B'^2_y - \alpha^2 \Rightarrow E'^2_x - B'^2_y = -3\alpha^2$$

$$\text{Solving } E'^2_x - B'^2_y = -3\alpha^2 \text{ and } E'_x + B'_y = \alpha \text{ we get } E'_x = -\alpha, \text{ and } B'_y = 2\alpha$$